

Employment effects of minimum wage indexing: Establishment evidence from Oregon restaurants

Stephen Miller¹ | Gary . Wagner²  | licia Plemmons³ 

¹Franciscan University of Steubenville,
Steubenville, Ohio, US

²B.I. Moody III College of Business,
University of Louisiana at Lafayette,
Lafayette, Louisiana, US

³John Chambers College of Business &
Economics, West Virginia University,
Morgantown, West Virginia, US

Correspondence

licia Plemmons.

Email: amplemmons@gmail.com

Abstract

Though 18 states will index their minimum wage to the Consumer Price Index by 2025, few studies have examined indexing's differential employment effects. Leveraging a period of stability in minimum wages (2000–2007) and two distinct national geocoded databases of establishments, we explore how indexing affected employment in Oregon restaurants, one of the earliest indexing states (2003). Nearest-neighbor matching is used as a preprocessing step before regression, pairing individual restaurants in Oregon to restaurants with similar characteristics in states where the minimum wage was unchanged. We find evidence that establishment employment falls 3.6% after indexing, implying an employment elasticity of -0.18 .

KEYWORDS

difference-in-differences, matching, minimum wage

JEL CLASSIFICATION

J08, J38, E31

1 INTRODUCTION

Eighteen states will index their minimum wage to the Consumer Price Index by 2025. However, relatively few studies seek to develop a benchmark for policy analysis by measuring indexing's effect on employment in periods of macroeconomic stability (Congressional Research Service, 2022). In anticipation of the upcoming rollout of inflation indexing in several states, including high-population states such as California and Florida, we leverage a period of relative stability in federal and state minimum wages (2000–2007) to estimate the effect of Oregon's indexing its minimum wage to inflation, relying on a national sample of firms matched on establishment-level attributes.

Indexing the minimum wage to inflation through the Consumer Price Index differs from the historical norm of ad hoc adjustments, as employers experience recurrent wage floor increases rather than a one-time policy shock. Therefore, employers must respond to an increase in the current wage and adapt their expectations to subsequent future increases (Brummund & Strain, 2020). Oregon was the second state (following Washington) to index its minimum

bbreviations: BDS, Business Dynamic Statistics; CBP, County Business Patterns; CIK, Central Index Key; D&B, Dunn & Bradstreet; GEOID, geographic identifiers; IPUMS, Integrated Public Use Microdata Series; LBD, Longitudinal Business Database; LLC, Limited Liability Company; N ICS, North American Industrial Classification System; NETS, National Establishment Time Series; NHGIS, National Historic Geographic Information System; QCEW, Quarterly Census of Employment and Wages; UWBDRC, University of Wisconsin Business Dynamics Research Consortium; YTS, Your Economy Time Series.

Managing Editor: David Sims

wage, which it did in 2003. The policy change occurred near the midpoint of a period of relative macroeconomic stability before the Great Recession that coincided with 21 states maintaining a minimum wage pegged to the federal rate of \$5.15 an hour. By 2007, Oregon's minimum wage had increased to \$7.80 because of incremental indexed changes, a 20% increase since the year before policy implementation. This period is also free from externally driven shifts in employment for the low-wage workforce, such as a pandemic or major financial disruptions, which may confound our understanding of how establishments respond on the employment margin.

The minimum wage literature in the United States has a deep-rooted methodological divide over whether it is more appropriate to observe the behavior of cross-border populations (Card & Krueger, 1994; Neumark & Wascher, 2000) or to use a case-study approach for single states or cities (Dube et al., 2010; Jha et al., 2022). We do not attempt to resolve this argument but provide an empirical framework that circumvents some of the limitations of each. That is, we compare individual establishments experiencing a systematic minimum wage change with matched establishments in states that retained the federal minimum wage. The focus on Oregon allows for a case-study analysis of one specific policy change. The matching procedure pairs restaurants sharing similar characteristics rather than assuming homogeneity among the aggregate establishments in a geographic or commuting zone.

Following the rationale of Brummund and Strain (2020), our analysis is limited to the restaurant industry, as this sector has a high proportion of minimum wage employees. Using both the National Establishment Time Series (NETS) and Your Economy Time Series (YTS) databases, we geocode individual establishment records and compile annual employment counts of all restaurants in the United States. Using single-nearest-neighbor and 1:k variable ratio matching to determine the closest comparison group(s), we then perform difference-in-differences estimation on the matched samples to isolate the causal effect of indexing.

cross numerous matched samples of restaurant establishments, we find that after indexing, Oregon restaurant employment fell, on average, 3.6% relative to their matched cohorts. These effects are consistent across multiple specifications and thousands of matching iterations to one or more restaurants. Restaurants are matched to other establishments of the same type (full-service, limited-service, cafeterias, buffets, etc.) that are located in census block groups sharing similar median household incomes and recent growth trajectories. Standard errors and confidence intervals are bootstrapped following badie and Spiess (2022).

Given that the nearly 3600 restaurants in Oregon entered the indexing period with an average of 16 employees, this 20% increase in the minimum wage resulted in an aggregate loss of more than 2000 jobs, implying an employment elasticity of -0.18 . While exhibiting a similar direction and significance, the disemployment effects we find during this time of relative economic stability are 60% smaller than the disemployment effects from indexing found by Brummund and Strain (2020). The methodological differences may partially explain this difference in effect size. While our methodology shares features with both the case study and the cross-border designs, it contributes to the minimum wage literature by presenting an alternative framework for analysis using establishment-level observations during a period of stable state and federal minimum wage rates.

Unlike Brummund and Strain (2020), who use industry-level data to examine the effect of indexing using both a contiguous county data set in addition to an all-county data set, we focus on matching establishments located in interior counties with less labor and consumer mobility as they are located further away from differently-affected states.

Additionally, Brummund and Strain (2020) compare employment outcomes under heterogeneous policy changes, where a subset of the states increasing minimum wage also indexed to inflation. This is a critical distinction as our study attempts to answer a similar but different question. We measure the effect of a state indexing its minimum wage relative to states that experience no changes in the minimum wage. This method develops a benchmark for policy analysis on the potential effects of indexing minimum wage if the alternative is to maintain the current policy. Our results indicate persistence in employment effects after indexing to inflation and, further, that these effects are more similar to other one-time shifts in the minimum wage than those shown in previous aggregated studies on indexing.

Our results provide valuable insights into the minimum wage policy discussion by identifying a period and design that facilitates analyzing the introduction of minimum wage indexing without large macroeconomic shocks that may destabilize labor markets. Our estimate of a -0.18 employment elasticity shows that in the absence of additional minimum wage changes or economic shocks, indexing the minimum wage reduces employment. These observations move beyond shared contiguous borders and instead match populations at the census block level based on economic factors, potentially yielding a more accurate comparison group for interior counties. The design also mitigates concerns of employees driving across state lines to access labor markets, which can inflate observed employment changes.

The paper proceeds as follows. Section 2 provides background on the current methodological debate in the minimum wage literature and details our contribution to the discussion of inflation indexing. Section 3 describes the NETS

and Your Economy Time Series databases, the matching procedure, and the empirical strategy. Section 4 discusses the results of the primary specifications and robustness of the results. Section 5 concludes.

2 BACKGROUND

There are frequent debates on the proper method for analyzing minimum wage changes and their effects on employment, earnings, and firm performance. While we do not attempt to resolve the disagreements, we employ a robust empirical approach that may circumvent some of the stated criticisms of both the cross-border and case-study approaches. We accomplish this by utilizing establishment-level data from two representative national samples during a period of relative economic stability.

Individual minimum wage changes raise wages by different amounts. They are implemented at different times in different cities and states. For these reasons, many economists find it appropriate to analyze them one at a time. The quasi-experimental case-study approach focuses on centralized geographic regions that share an employment pool, are bifurcated by state or county borders, and differ only in minimum wage policies. These studies have risen in popularity since the beginning of differencing models that Card and Krueger (1994) compared employment levels in fast-food restaurants in New Jersey and Pennsylvania following an increase in New Jersey's minimum wage in 1992. This study, which marked the beginning of a comparative focus in the literature, has since become emblematic of minimum wage research.

Much of the research on the employment effects of the minimum wage has focused on the workers most likely to be impacted by increases, including teenagers, adult high-school dropouts, restaurant employees, hotel employees, and retail employees. These workers are likely to be disadvantaged by reduced employment opportunities and at a greater risk of poverty from loss of wages in low-income occupations. Quasi-experimental literature commonly uses survey and observational data, such as help-wanted ads for hospitality workers in Oregon and Washington (Singell & Terborg, 2007), phone surveys of managers at popular fast food chains like Burger King and Kentucky Fried Chicken (Card & Krueger, 1994; Katz & Krueger, 1992), Bureau of Labor Statistics establishment-level data (Card & Krueger, 2000), and payroll data (Neumark & Wascher, 2000), consistently finding either negative or no effect on employment rates.

Early minimum wage research utilized the National Longitudinal Survey of Youth, comprising predominantly individual-level cross-sectional data sources, to estimate the wage changes for workers affected by minimum wage increases (Currie & Fallick, 1996). As the literature has moved from case studies to broader border discontinuity designs, data aggregated at local and state levels have become more prominent. Card and Krueger (1994) initially estimated the employment elasticity of teenagers in response to changes in the federal minimum wage using the Census Bureau's Current Population Survey. Other studies expanded and supplemented this work with aggregate data collected from the Census Bureau's Quarterly Census of Employment and Wages (QCEW) to analyze earnings discontinuities at state borders (Dube et al., 2010), effects on state-specific teenage employment (Card, 1992), and employment patterns in commuting zones (Jha et al., 2022).

Several challenges persist in the minimum wage literature. For example, the size of the minimum wage change can influence the duration and magnitude of potential employment effects in the short and medium term (Clemens & Strain, 2021). Powers et al. (2007) examine an increase from \$5.15 (at the time, the federal minimum wage) to \$5.50 per hour in Illinois in 2004, then a subsequent increase to \$6.50 in 2005. Over these 2 years, the \$1.35 change represented a nearly 26 percent increase in the wage rate, greater than the more modest \$0.80 (19 percent) increase in the New Jersey study performed by Card and Krueger (1994). Ultimately, the authors disagreed about the appropriate interpretation of the results and whether they could be used to theorize different-sized minimum wage changes (Persky & Baiman, 2010; Powers, 2009).

Another additional challenge in the minimum wage literature is the duration of real wage and employment effects, which macroeconomic fluctuations may confound. Because most minimum wage increases are small, one-time policy adjustments, inflation eventually erodes their effects. Historically, federal and most state-specific minimum wages have not kept pace with inflation (Bivens, 2022). However, suppose the introduction of capital investment and automation makes labor demand elasticities higher in the long run than in the short run. In that case, reductions in employment may be observable for some period after a minimum wage increase.

Recently, states have addressed the inability of the minimum wage to meet employees' changing purchasing needs by raising the minimum wage above the federal requirement and indexing it to inflation. As of 2023, 18 states and the

District of Columbia have authorized or implemented policies to index their state's minimum wage to inflation each January. The inflation rate is frequently assigned to the August-to-August changes in the consumer price index during the previous year (Congressional Research Service, 2022).

Of these states, the overwhelming majority approved and passed the inflation indexing through ballot measures between 2014 and 2018, with the scheduled initial adjustments beginning in the early to mid-2020s. The largest cohort of states, including Alaska and California, recently announced their first inflation-adjusted increases effective January 1, 2023. These inflation-indexed minimum wage changes are unique in that firms anticipate an increase in the short-term and also additional annual increases of variable amounts depending on economic conditions. These long-run increases negate the nominal increase effect. In other words, because inflation will not rise above the minimum wage, minimum wage increases will be binding in the long term.

Numerous factors affecting the purchasing power of low-income workers will make the effects of the minimum wage policy change difficult to evaluate in states that introduced indexing in the early 2020s. These include the pause on student loan payments, expansions to Medicare and Medicaid, and aggressive interest rate hikes by the Federal Reserve. The study closest to our own, Brummund and Strain (2020), focuses on the few states that indexed before 2020 and analyzes the duration of employment effects within the restaurant industry between 1990 and 2016. That study finds heterogeneous effects of indexing, with disemployment effects roughly three times the magnitude of a one-time nominal increase in the minimum wage. Even though the indexed policy continues increasing the minimum wage into perpetuity, the shock appears to be transitory, with disemployment effects persisting for approximately 14 quarters after implementation (Brummund & Strain, 2020).

We complement the Brummund and Strain (2020) study in at least four dimensions. First, we remove the confounding effects of the Great Recession by focusing on Oregon during a period of relative macroeconomic stability: 2000–2007. Brummund and Strain (2020) include this period of relative macroeconomic stability, but also have several potential confounding effects between 1990 and 2016, preventing us from disaggregating their results only during our period of interest. Second, instead of focusing on data aggregated to the county, county pairs, or state, which other local economic policy changes may influence, we utilize establishment-level data on restaurants within Oregon and match these establishments against a representative national sample using individual restaurant and local socioeconomic-specific attributes. Third, we disregard border counties with shared labor and consumer markets and instead focus on the within-establishment effect between matched establishments located in interior counties. Focusing on matched interior counties allows for less labor and consumer mobility to differently affected states, resulting in lower employment elasticity. Fourth, we analyze the employment effects of introducing an indexed minimum wage when the alternative is to maintain the status quo, rather than differences across heterogeneous policy changes where a subset introduces indexing, and provide insight into the employment effects of indexing in the absence of competing policy changes.

Similar approaches using a period of relative economic stability between 2012 and 2015 have also been utilized by Clemens and Strain (2018) to analyze employment effects among low-skilled populations in the American Community Survey and Clemens and Strain (2017) for documenting education and employment changes for young adults.

Oregon is a unique case in that it indexed its minimum wage near the midpoint of a period of relative economic stability when few other states were changing minimum wage policies. Oregon's minimum wage diverged from the federal rate in 1990 when it was raised to \$4.25 (and the federal rate was \$3.25). In the late 1990s, Oregon increased its minimum wage for three consecutive years, increasing it from \$0.75 to \$5.50 in 1997, to \$6.00 in 1998, and finally to \$6.50 in 1999. The state's rate remained constant (in nominal terms) until 2003, when it was raised to \$6.90, and indexing became effective.¹ The choice to index the minimum wage came after a citizen-initiated ballot measure that amended the state statute, known as Oregon Measure 25, which had overwhelming resident support and was approved in November 2002. By 2007, the end of this period of relative stability, Oregon's minimum wage had reached \$7.80, an increase of 20 percent from 2002.

Previous studies have struggled to resolve the debate on the most appropriate treatment and control groups. Dube et al. (2010) advanced the minimum wage literature by generalizing what was historically a case-study approach to the entire nation on the dual assumptions that economic activity is shared across state boundaries and that local economic factors need not be considered. Their analysis attributes the differences in employment effects to spatial heterogeneity and finds that minimum wage increases have little effect on employment. At the same time, this contiguous county approach allows for comparison of employment among groups sharing similar geographic and regional features (e.g., Card & Krueger, 1994; Dube et al., 2010; Powers et al., 2007; Singell & Terborg, 2007), some economists have argued that these comparison populations may still differ in other ways such as demographics or non-cash compensation

policies (Clemens, 2021), and have attempted to correct for these potential forms of bias by incorporating commuting zones (Jha et al., 2022) or synthetic control populations (Powell, 2022).

Our matching procedure, using establishment-level data from the interior of the states, circumvents the issues that populations sharing a border may differ in local-level economic policies and cross-border mobility of workers and firms by matching each restaurant to a representative comparison restaurant of the same type that shares similar local-level trend economic activity. Rather than comparing Oregon counties to contiguous counties in neighboring states, we use national data to compare establishment-level employment in Oregon restaurants with restaurant employment from a large pool of states that did not change their minimum wages between 2000 and 2007. As such, this paper adds to the literature on minimum wage indexing and uses establishment-level restaurant employment to estimate labor demand elasticities.

3 DATA AND EMPIRICAL SPECIFICATION

3.1 Data

We utilize two private-sector databases for establishment-level employment in restaurants, the NETS and YTS databases. While each database is created from proprietary tools and processes, discussed in more detail below, both aim to capture the universe of business establishments operating in the U.S. In addition to employment counts at the establishment level, both NETS and YTS provide additional information on individual establishments, including latitude and longitude and the primary North American Industrial Classification System (NAICS) at a six-digit level, among other characteristics.

The NETS database, created by Walls & Associates, LLC, is derived from information collected in Dunn & Bradstreet's (D&B) Duns Marketing Information files (Barnatchez et al., 2017). D&B tracks individual establishments from public sources of information such as court records and secretaries of state, as well as direct contacts by telephone, to create data products for other businesses (Barnatchez et al., 2017). The NETS database, available since 2003, provides annual information on establishments (back to 1990) collected from Dunn & Bradstreet's snapshot files from January of the calendar year of interest (Walls & Associates, 2019).

Similarly, the YTS database is a collaborative effort between Data Xle and the University of Wisconsin Business Dynamics Research Consortium. Like Dunn & Bradstreet's, Data Xle monitors business activity to provide data solutions to other businesses. Stated goal of Data Xle is to identify establishments that are or intend to be "in-business" from establishments that exist only to hold assets or otherwise do not engage in commercial activities. The public data product, available from the University of Wisconsin, includes additional geographic information (congressional district, etc.) and information such as the Security and Exchange Commission's central index key to facilitate linking publicly traded companies with other datasets. YTS is a newer data product than NETS (first available in 2015) and provides annual coverage back to 1998.²

While both the NETS and YTS databases have been used in numerous studies, they have yet to be used to study the minimum wage. Prominent examples of studies using NETS includes Rossi-Hansberg et al. (2018), Groizard et al. (2015), Neumark et al. (2011), and Neumark and Kolko (2010). Recent examples using YTS have focused on Amazon distribution centers (Houde et al., 2023), corporate real estate activity (Campello et al., 2022), and opportunity zones (Coyné & Johnson, 2023).

Given its longer existence, NETS has been compared more extensively to other data sources. For instance, Barnatchez et al. (2017) compare NETS to several government data sources, including the County Business Patterns (CBP), QCEW, and non-employer Statistics, and find some notable differences. First, while the NETS data generally mimic data found in other sources, there is more imputation in the NETS data, particularly among small establishments. At a sectoral level, Barnatchez et al. (2017) find the largest discrepancies in education services (NAICS 61).

While Barnatchez et al. (2017) focus on aggregate (national) employment counts, recent work by Zeballos and Marchesi (2022) compares NETS and CBP in food-related industries at a national and subnational level. In the restaurant sector (NAICS 7225)—the focus of this study—Zeballos and Marchesi (2022) find that aggregate employment counts between NETS and CBP differ by an average of only 1% from 1997 to 2016. The close national relationship also holds at the state level (see fig. 7 in Zeballos & Marchesi, 2022).³

The NETS and YTS databases have other similarities and differences with official government data sources that are worth noting. For instance, NETS and YTS share the same employment concept as the Census Bureau's restricted-use

Longitudinal Business Database, QCEW, CBP, and Business Dynamic Statistics programs. Employee counts include full-time workers, part-time workers, and temporary workers at the time the survey was conducted. A key difference between NETS, YTS, and official government data sources, including the Economic Census (conducted every 5 years), is that NETS and YTS include owners in employee counts and non-employer establishments. To the best of our knowledge, non-employer establishments are excluded from all official government measures, so “businesses” are restricted to establishments that have at least one paid worker at the time of data collection regardless of whether the worker is full-time, part-time, or temporary.

Additionally, the NETS and YTS databases differ from the QCEW and CBP along other dimensions. First, the QCEW is derived from (government-defined) establishments that are part of either state or federal unemployment insurance programs. There is variation in state exemptions, with the most common being establishments in agricultural activities or some businesses on tribal lands. County Business Patterns, which is based on the Census Bureau's Business Register, has broader coverage on tribal lands than the QCEW but also excludes some industries, most notably agriculture and rail transportation. Government-run establishments, regardless of location, are excluded from both QCEW and CBP.⁴ While NETS and YTS may have an advantage in the sense that both databases strive to be more inclusive than official sources, the data are reported at an annual frequency so they are not ideally suited for examining short-run dynamic responses to policy shocks.

Despite the methodological differences, comparing NETS and YTS to official data sources like the QCEW and CBP is useful. Figure 1 provides a comparison for the interior (or non-border) counties in Oregon during the sample period (2000–2007). Our following empirical analysis is limited to interior (or non-border) counties because there is evidence that low-wage jobs move across borders in response to minimum wage increases (Pérez, 2022). Other state or local policies could also contribute to job flow across borders, so there are reasons apart from minimum wage changes to exclude border counties.

Panel a in Figure 1 compares (raw) establishment counts in the restaurant industry in Oregon during our sample period for NETS, YTS, QCEW, and CBP.⁵ Across the sample period, no data source consistently reports either the highest or lowest establishment counts. QCEW reports a smaller number of establishments every year except in 2001, when the YTS database reported the fewest restaurants. On the high end, NETS reports the highest establishment counts in 5 years (2001, 2002, 2005, 2006, 2007), YTS reports the highest counts in 2 years (2003, 2004), and CBP reports the highest count in 2000.

Over the 8-year period, the median difference between the highest reported establishment count and lowest reported establishment count is (in absolute value) 9.7%. Compared to the QCEW and CBP, the median percentage difference from NETS (in absolute value) is 8.4% and 4.0%, respectively. YTS establishment counts, in contrast, tend to be closer to both QCEW and CBP counts. In percentage terms, the median difference (in absolute value) is 4.0% between YTS and CBP, and 3.3% between YTS and QCEW. The primary difference between NETS and YTS is that NETS reports higher counts of non-employer (or owner-only) start-up establishments.

In terms of trends, establishment counts are trending upwards across all four datasets during the sample period (Panel a of Figure 1). Specifically, between 2000 and 2007, counts of restaurants in Oregon expanded by 27.6% according to the QCEW, 28.3% according to CBP, 46.3% according to NETS, and 29.5% according to YTS. The higher counts of non-employer establishments in NETS, particularly start-ups in 2006 and 2007, are the main drivers of the higher growth rate observed in NETS over the sample period. It is infeasible to know the exact sources of discordance between NETS and YTS without specific knowledge of the proprietary collection and validation processes involved in producing each data product. However, we believe that using both NETS and YTS provides an attractive robustness check to corroborate or rebut any findings.

Panel b in Figure 1 compares average establishment employment formed from aggregate employment and establishment counts. On average, the QCEW reports the highest mean employment count per establishment at 16.0, followed by CBP (15.4), YTS (15.1), and NETS (14.4). Similar to raw counts of establishments, the inclusion of non-employer establishments in NETS and YTS has the effect of reducing average employment counts. If one omits non-employer establishments (those reporting a single employee) from both datasets, then the mean employment count (per establishment) in NETS and YTS becomes 15.1 and 15.6, respectively. Considering Panels a and b in Figure 1 jointly, both the NETS and YTS datasets seem to generally track official government counts of restaurants and per-establishment employment in Oregon's interior counties during the period of interest.

With border counties being omitted, the empirical analysis compares restaurants located in interior (or non-border) counties in Oregon to restaurants located in 1121 interior counties in the 21 states where the nominal minimum wage was unchanged.⁶ Idaho is Oregon's only neighboring state to have a constant (nominal) minimum wage during the



FIGURE 1 Comparison of NETS and YTS to official government sources: Oregon. This figure compares the total establishment count (top panel) and the average number of employees per establishment (bottom panel) in the restaurant industry in Oregon, including border counties. As noted in the text, both the NETS and YTS databases include non-employer establishments, which are both excluded from CBP and QCEW. NETS and YTS use the 2017 version of N ICS, whereas CBP and QCEW are based on the 2002 and 2007 versions. Figures for CBP and QCEW are from (2002 or 2007) N ICS 7221 and 7222, which are the equivalent to 7225 N ICS in 2017. CBP, County Business Patterns; QCEW, Quarterly Census of Employment and Wages; NETS, National Establishment Time Series; YTS, Your Economy Time Series.

sample period. California increased its minimum wage in 1999, 2001, 2002, and 2007. Nevada increased its minimum wage in 2007, and Washington's minimum wage increased annually beginning in 1999 due to indexing.

The states included and excluded from the empirical sample are shown in Figure 2. The numbers shown on each state are the number of distinct restaurants located in interior (or non-border) counties that are observed at least once in both the pre (2000–2002) and post (2003–2007) treatment periods from the NETS and YTS databases. Counts from the NETS database are reported as the top figure, while restaurant counts from YTS are shown as the bottom figure in parentheses.

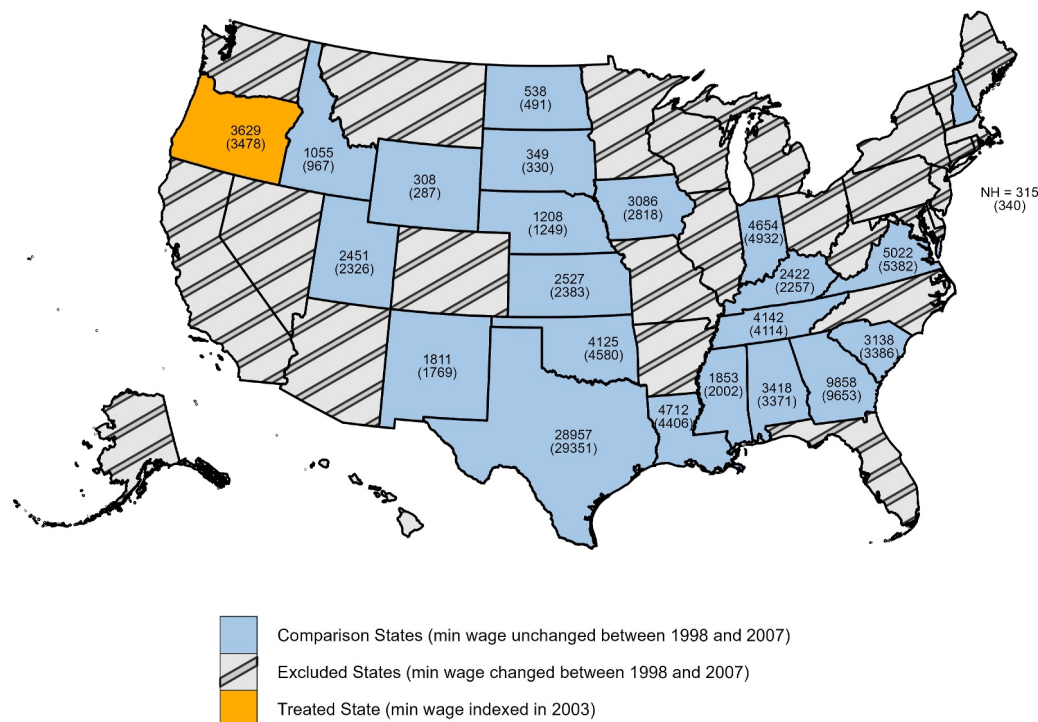


FIGURE 2 Number of restaurants in treated and comparison states. Figures for each state show the number of non-border county restaurant establishments (N ICS 7225) that are observed at least once in the pre (2000–2002) and post (2003–2007) treatment periods from the NETS database. Figures in parentheses are counts, based on the same criteria, from the YTS database. The treatment period is 2003, when Oregon indexed its minimum wage to the Consumer Price Index. The comparison states all had stable (nominal) state minimum wages between 1998 and 2007 that were equal to the federal minimum wage of \$5.15. States excluded from the empirical sample, shown with the stripes, changed their state minimum wage at least once between 1998 and 2007. *Source:* Vaghul, Kavya and Zipperer, Ben (2016). “Historical State and Sub-State Minimum Wage Data,” Washington Center for Equitable Growth (<https://equitablegrowth.org/working-papers/historical-state-and-sub-state-minimum-wage-data/>). NETS, National Establishment Time Series; YTS, Your Economy Time Series.

According to the NETS database, there are a total of 6401 restaurants in Oregon that operated before and after indexing, with 3629 located in interior counties and the remaining 2772 located in border counties. There is a similar pattern from the YTS database, with 6246 restaurants operating before and after indexing. Of these establishments, 3478 are located in interior counties and the remaining 2768 are located in border counties. Multnomah County, which borders Washington and contains the City of Portland, accounts for 52.3% of all border county restaurants in NETS and 53.9% of all border county restaurants in YTS. Jackson County, which borders California, is home to Oregon’s second largest share of border county restaurants with 14.1% (NETS) and 13.3% (YTS), respectively.

The difference-in-differences sample of restaurant counts discussed in the preceding paragraph differs from the raw establishment counts reported in Panel a of Figure 1. This is because the difference-in-differences analysis is restricted to establishments that we observe at least once before and after indexing. Hence, start-up restaurants observed for the first time after indexing took effect in 2003, which are more prominent in the NETS database, appear in Panel a of Figure 1 but are omitted from the regression analysis.⁷

Shifting to comparison states in Figure 2, NETS and YTS report a total of 85,949 and 86,394 interior restaurants that operated before and after indexing. NETS reports higher establishment counts in all states except Indiana, Mississippi, Nebraska, New Hampshire, Oklahoma, South Carolina, Texas, and Virginia. In percentage terms, NETS and YTS report the closest figures in Tennessee (a differential of 0.68%) and the largest discrepancy in Oklahoma (a differential of 9.9%).

Across all 22 states in the sample, the mean and median differential in restaurant counts between NETS and YTS in absolute value is 5.7% and 6.3%, respectively.

For the states and counties in the sample, the distribution of establishment sizes is very similar for NETS and YTS, as shown in Panel a of Figure 3. For example, the fraction of (non-border county) restaurants that always report a single employee during the sample period is 2.8% in NETS and 2.4% in YTS. If one broadens the definition to establishments that ever report a single employee during the sample period, then the shares increase to 7.6% in NETS and 6.7% in YTS.

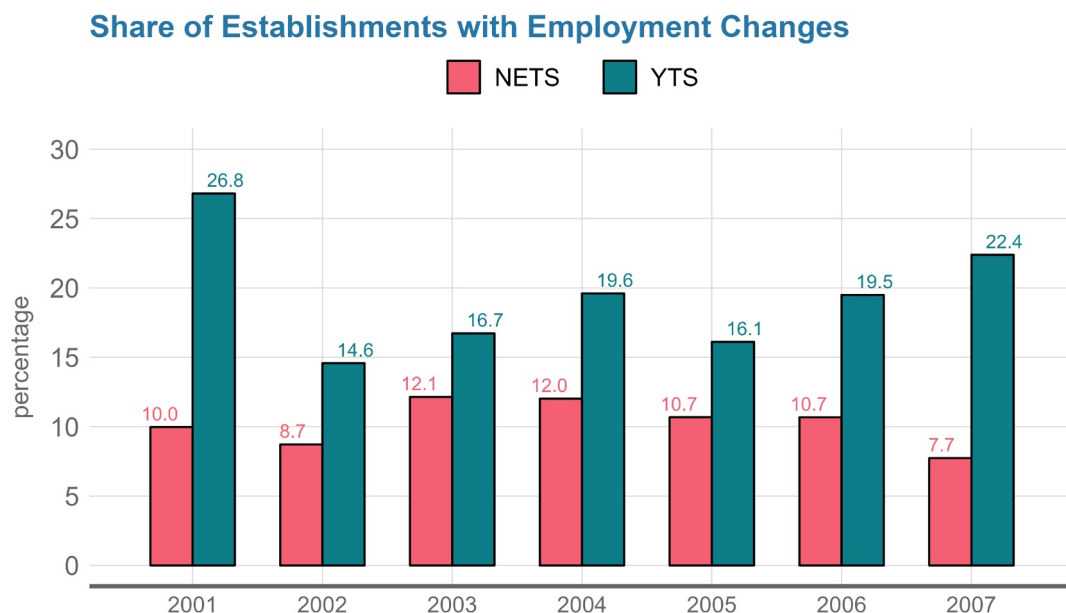
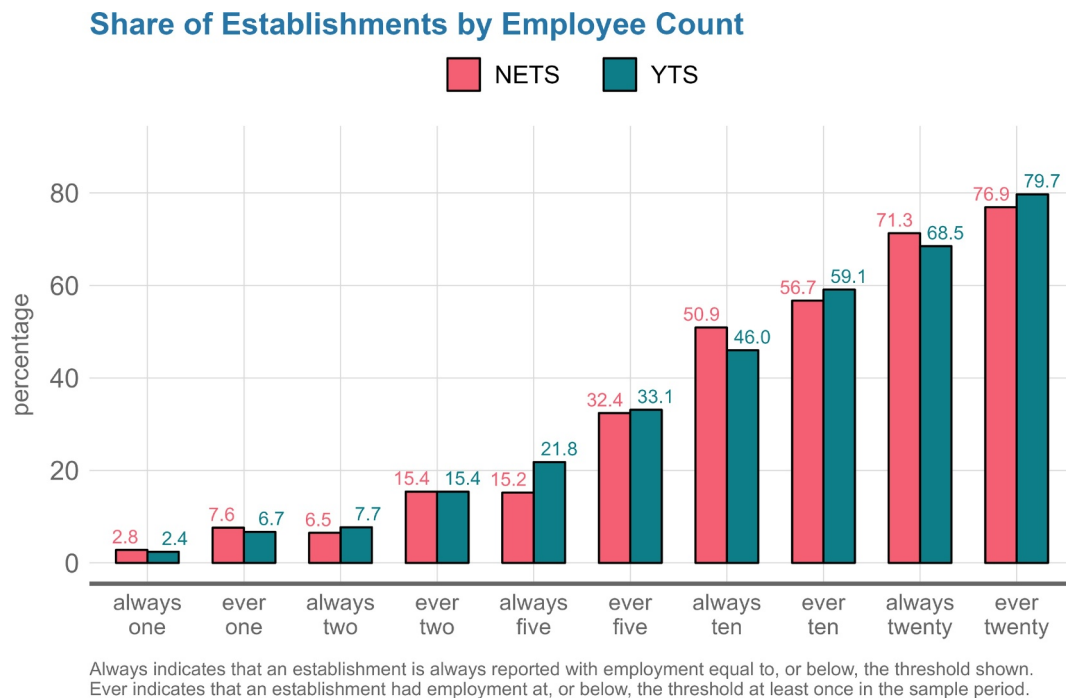


FIGURE 3 Distributional comparison of NETS and YTS establishments: sample counties. NETS, National Establishment Time Series database; YTS, Your Economy Time Series database.

Almost 16% of establishments in both datasets report two (or fewer) employees at some point during the sample period. One-third of all restaurants in the sample are recorded as having 5 (or fewer) employees at least once between 2000 and 2007. As a robustness check, we also redo our analysis by: (i) excluding establishments that always report a single employee, and (ii) excluding establishments that always report two (or fewer) employees.

More generally, approximately half of all restaurants in NETS and YTS samples always report having 10 or fewer employees. These percentages grow to almost 60% for restaurants that, at any point in the sample, have reported employment counts that ever reach, or fall below, 10 workers. Fewer than 20% of establishments always report more than 20 workers, illustrating how much small-employment establishments dominate the industry. An examination of the names of these “large” restaurants that always report 20 or more workers suggests that they are dominated by national chains. In terms of distributional comparisons, Barnatchez et al. (2017) compare NETS and imputed County

Business Patterns after excluding establishments with 2 employees, 5 employees, and 10 employees. They find that the percentage differences between datasets, measured by total employment at the state level, remains largely unchanged, suggesting similar distributions.⁸

While the distribution of establishments by employee counts is quite similar in NETS and YTS, there is more variation in year-to-year employee counts reported by the datasets. Panel b in Figure 3 shows the fraction of all establishments experiencing year-over-year employment changes. On average across the sample, NETS reports that an average of 10.3% of establishments undergo staffing changes, whereas YTS reports almost twice as much annual employment churn with an average of 19.4%.⁹

3.2 Matched sample formation

conventional difference-in-differences framework comparing (non-border county) restaurants in Oregon to all the (non-border county) restaurants in the comparison states revealed “pre-trends” violations. Significant pre-treatment differences between the treated and comparison groups persisted when using a doubly robust approach (Sant’ nna & Zhao, 2020).

Given that the number of comparison establishments greatly exceeds the number of treated establishments in both datasets, we construct several matched samples of restaurants (prior to estimating regressions) to better align treated and comparison restaurants and reduce potential sources of bias. Creating matched samples in the preprocessing or design phase before regression, initially proposed by Rubin (1973), is a common strategy to reduce bias in observational studies (badie & Spiess, 2022; Ho et al., 2007).¹⁰

matched sample combined with regression shares a similarity with doubly robust estimators in the sense that observations are weighted. There is, however, a key difference. Doubly robust estimators, such as badie (2005) and Sant’ nna and Zhao (2020), include all treated and comparison units and weight units using some variation of their inverse propensity score. Units (treated or comparison) that have dissimilar covariate distributions remain in the sample, which may lead to greater reliance on extrapolation and, ultimately, bias (Imbens & Rubin, 2015). The estimators are doubly robust because consistency is achieved if either the propensity score or outcome model is correctly specified (Sant’ nna & Zhao, 2020). matched sample, or trimming approach, seeks to improve the covariate balance between treated and comparison units *prior* to estimation (Imbens & Rubin, 2015). If there are treated or comparison units that fall outside the support of the data, trimming these units can reduce dependence on extrapolation and lead to more credible estimates (Imbens & Rubin, 2015).

We leverage the detailed geographic nature in NETS and YTS to match restaurants in Oregon to restaurants in comparison states that share similar (pre-treatment) local characteristics and economic trajectories that could explain trend differences in restaurant employment before indexing was implemented. Drawing on data from the 1990 and 2000 Censuses, we perform (cross-sectional) propensity score matching via logistic regression for our restaurant samples from NETS and YTS. We estimate separate propensity scores for each dataset, with treated and untreated restaurants appearing once in each regression. This results in a sample size of 89,572 (3629 treated) restaurants using the NETS database and a sample size of 89,872 (3478 treated) restaurants using the YTS database. The propensity score equation can be formally expressed as:

$$\Pr(T = 1|\mathbf{Z}) = \frac{\exp(\mathbf{Z}'\beta)}{1 + \exp(\mathbf{Z}'\beta)} \quad (1)$$

where T denotes the treatment indicator (1 = treated; 0 = untreated) and $\mathbf{Z}$ is a set of observed covariates. The propensity score (or predicted value from the logistic regression) for an individual unit, $e_i(x_i)$, is the probability that a restaurant is assigned to the treatment group conditional on covariates (Rosenbaum & Rubin, 1983). This matching step allows us to identify untreated restaurants outside of Oregon that look very similar (or even identical) to treated restaurants in Oregon based on the economic and establishment characteristics used in estimating the propensity scores.

s Rosenbaum and Rubin (1983) and Rubin and Thomas (2000) note, if treated and comparison establishments have the same propensity score, then these two restaurants will have identical distributions of covariates used in the propensity score estimation. The objective of this type of “design phase” matching is to reduce the imbalance in the distribution of non-outcome observable characteristics between treated and untreated units (Imbens & Rubin, 2015). Since (observable) county- or state-level data that could affect restaurant employment trends can easily be incorporated

into a conventional regression framework, we perform the matching step using census block group characteristics that are unavailable outside of decennial census years and cannot be integrated into a regression equation. The objective of the matching step is to balance on local economic forces that may explain differential pre-trend employment levels between restaurants in Oregon and restaurants in other states to construct a more credible counterfactual for Oregon's restaurants.

For both the NETS and YTS databases, the propensity score equation predicts the probability of treatment (being located in Oregon) from: (i) four indicator variables for the restaurant's subclassification within the N ICS hierarchy (full-service—722,511, limited-service—722,513, cafeterias and buffets—722,514, and snack and nonalcoholic beverage bars—722,515), (ii) median household income in the restaurant's census block group in 2000, (iii) the average annual growth rate in a restaurant's census block group population between 1990 and 2000, (iv) the average annual growth rate in a restaurant's census block group median house value between 1990 and 2000, (v) the average annual growth rate in a restaurant's census block group poverty rate between 1990 and 2000, and (vi) the average annual growth rate in a restaurant's census block group number of housing units between 1990 and 2000.¹¹

Indicator variables for the restaurant's N ICS subclassification will adjust for differences across establishment types. If a full-service restaurant requires more employees than other restaurants and the composition of restaurant types varies across states, potentially due to regulatory differences, then restaurant type could be correlated with employment outcomes. In addition, (block group) median household income in 2000 will capture the general level of economic well-being in a restaurant's local market at the start of the sample period. If else equal, restaurants located in more affluent areas may experience more traffic and require additional employees. Finally, the average annual growth rates in population, poverty rate, median home value, and number of housing units over the previous 10 years will capture longer-term economic trends in a restaurant's local market that could explain trend differences in establishment-level employment, all else equal.

Establishment employment counts are omitted from the matching step for several reasons. First, Imbens and Rubin (2015) recommend excluding the outcome variable from the matching process in order to avoid deliberately biasing final estimates (p. 283). Additionally, matching on historical employment trends is complicated by the fact that states have made frequent changes in their minimum wage policies which could affect trend growth. While we could match on the level of employment in 2000, which Imbens and Rubin (2015) recommend against, it is also possible that two restaurants with identical employment counts could be trending in different directions.

In our application, a treated and untreated restaurant that have “similar” propensity scores will—on average—be the same type of restaurant (full-service, limited-service, etc.), be located in a local market with similar median household incomes, and have similar long-run trends in local population, poverty, and housing markets.¹²

We follow the matching strategy outlined in Rubin and Thomas (2000). Matches for treated units occur sequentially, starting with the unit having the highest propensity score (the most difficult to match). Comparison units are considered potential matches if their propensity score falls within some maximum allowable difference (in absolute value) from the treated unit, referred to as a caliper. If no comparison units fall within the caliper, then the treated unit is unmatched and discarded from the sample. If multiple comparison units are within the caliper, the one with the closest propensity score is considered a match. Matching is done without replacement, so a comparison restaurant is matched to a single treated unit. As a robustness check, we also match in random order (explained in more detail in Section 4.1).

Rubin and Thomas (2000) recommend a caliper of 0.02 times the standard deviation of the propensity scores. We examine robustness using calipers of 0.03, 0.01, and 0.005, referred to as wide, mid, and narrow. Wide calipers allow for more matches but raise concerns that matches may be insufficiently close. Narrower calipers ensure closer propensity scores, thereby lowering potential bias, but may also increase the likelihood that more treated units will go unmatched.

For each caliper, we also use single-nearest-neighbor matching and 1:k variable ratio matching. The single-nearest-neighbor match ensures each matched restaurant is paired with its closest comparison restaurant. While expanding the number of potential matches may introduce bias by including comparison units with more dissimilar propensity scores, Rosenbaum (2020) has shown that variable ratio matching can further reduce bias in some settings. Since the benefit of variable ratio matching falls once k exceeds 5 (Rosenbaum, 2020), we match each treated establishment up to a maximum of 5 comparison establishments as a robustness check.¹³ One distinct advantage of variable ratio matching is that treated restaurants are compared to broader, and different, sets of comparison restaurants. This provides a natural robustness check.

To quickly summarize, suppose we are examining all establishments, including non-employer establishments, and matching in sequential order (hardest to match first). For this “scenario,” we estimate a total of 12 regressions—six

using the NETS dataset and six using the YTS dataset. The six regressions per dataset use nearest neighbor and variable ratio matching to form comparison groups for each of the three calipers: wide, mid, and narrow.

It is important to note that the estimand changes when treated units are discarded, so matching as a preprocessing step before regression generally does not recover the average treatment effect on the treated (Imbens & Rubin, 2015). We are instead recovering the causal effect of indexing the minimum wage for the subpopulations in each of the matched samples.¹⁴

Difference-in-differences could be performed without regression on the matched samples if one assumes strong ignorability, meaning that all possible confounders are included in the propensity score. Unfortunately, many highly localized characteristics that one might expect to affect restaurant performance, such as zoning requirements, crime rates, or city ordinances pertaining to food safety, are not readily available, so they could not be included in the matching process. For this reason, we include (county or state-level) time-varying covariates and establishment fixed effects to adjust for these characteristics. The regression specification is given by:

$$\log \text{employment}_{jst} = \alpha + \delta D_{jst} + \beta \mathbf{X}_{cst} + \sigma_j \mathbf{E}_j + \theta_t \mathbf{Y}_t + \epsilon_{jst} \quad (2)$$

where $\log \text{employment}_{jst}$ denotes the natural log of the number of employees in establishment j , located in county c and state s at time t . D_{jst} is the unit-time treatment indicator that equals one for the treated establishments in Oregon beginning in 2003 and equals zero otherwise. δ is the coefficient of interest.

$\mathbf{E}_j$ and $\mathbf{Y}_t$ denote establishment and year fixed effects, respectively. The year fixed effects will adjust for common shocks to all establishments. $\mathbf{X}_{cst}$ represents a vector of county-level and state-level time-varying controls, including median home prices, per capita persona income, gasoline prices, and public finance variables.¹⁵

At the county level, we include per capita personal income (in \$000,000s) to adjust for differences in the average level of economic well-being. We also include median home prices to adjust to potential differential wealth effects resulting from the housing boom of the mid-2000s. Home price appreciation, and borrowing against home equity, have been directly linked to consumption patterns during our sample period (Belsky & Prakken, 2006). There is also evidence that different regions of the country experienced differential housing booms and busts based, in part, on local characteristics such as land use restrictions, mortgage underwriting standards, the share of second homes, and the share of subprime borrowers (García, 2022; Griffin et al., 2021; Nathanson & Zwick, 2018). Home prices are measured in millions of dollars.

The remaining time-varying covariates are measured at the state level. We include the effective overall state tax rate, effective personal income tax, and effective property tax rate to adjust for differences in how states finance state and local services. Effective tax rates are defined as total tax revenue collected as a fraction of state personal income. Oregon is one of only five states without a statewide general sales tax (Advisory Commission on Intergovernmental Relations, 1994). The state relies, instead, more heavily on personal income taxes to finance state services and local property taxes to finance local services (Advisory Commission on Intergovernmental Relations, 1994). During our sample period, Oregon had the highest mean effective personal income tax rate (4.1%) among states in the sample. Since unincorporated establishments fall under the personal income tax structure, local tax structures are potentially important factors to consider.

The final time-varying covariate we include is the statewide annual average retail gasoline price, measured in cents per gallon. There is evidence that higher gas prices both reduce consumer confidence (Binder & Makridis, 2022) and influence how frequently people eat at restaurants (Courtemanche, 2011). Retail gasoline prices in Oregon also tend to be among the highest in the nation because of prohibitions on self-service stations, higher than average taxes, and a lack of refining capacity (General Accountability Office, 2001). During our sample period, Hurricanes Katrina and Rita led to sizable differential retail gas price shocks around the nation, some of which were (at the time) nominal highs (Blair & Rezek, 2008).

Data source information and descriptive statistics for raw (unweighted or unmatched) and matched (or weighted) samples are reported in Table 1.¹⁶ NETS sample descriptive statistics are reported in Panel A, while the YTS descriptive statistics are reported in Panel B. In Table 1, matching weights were obtained using sequential order matching using the widest caliper, which matches the most difficult establishments to pair first.

In the unmatched (or raw) data shown in columns (1) and (2), restaurants in Oregon have an average of 15.40 employees, compared to 18.37 in the comparison states using the NETS database. Using the YTS sample, treated and comparison restaurants have average employment counts of 15.92 and 18.12, respectively. In the unweighted samples,

TABLE 1 Descriptive statistics: matched and unmatched samples.

Variable	(1) Treated unmatched	(2) Comparison unmatched	(3) Treated wide caliper	(4) Comparison max 1 NN	(5) Comparison max 5 NN	(6) Treated mid caliper	(7) Comparison max 1 NN	(8) Comparison max 5 NN	(9) Treated narrow caliper	(10) Comparison max 1 NN	(11) Comparison max 5 NN
Panel A : National Establishment Time Series (NETS)											
Number of employees	15,408 (18,698)	18,376 (35,942)	15,463 (18,747)	15,363 (22,961)	15,967 (28,789)	15,509 (18,749)	15,297 (22,971)	15,891 (29,142)	15,541 (18,707)	15,307 (23,164)	15,785 (28,090)
Median home price	0.173 (0.054)	0.121 (0.045)	0.173 (0.054)	0.126 (0.045)	0.127 (0.045)	0.173 (0.054)	0.126 (0.045)	0.127 (0.045)	0.173 (0.054)	0.126 (0.045)	0.127 (0.045)
Gasoline price	1.564 (0.511)	1.420 (0.500)	1.564 (0.511)	1.429 (0.496)	1.429 (0.496)	1.564 (0.511)	1.429 (0.496)	1.429 (0.496)	1.563 (0.511)	1.430 (0.496)	1.430 (0.496)
Effective total tax rate	0.058 (0.004)	0.056 (0.010)	0.058 (0.004)	0.060 (0.010)	0.060 (0.010)	0.058 (0.004)	0.060 (0.010)	0.060 (0.010)	0.058 (0.004)	0.060 (0.010)	0.060 (0.010)
Effective income tax rate	0.041 (0.003)	0.014 (0.012)	0.041 (0.003)	0.018 (0.011)	0.018 (0.011)	0.041 (0.003)	0.018 (0.011)	0.018 (0.011)	0.041 (0.003)	0.018 (0.011)	0.018 (0.011)
Effective sales tax rate	0.000 (0.000)	0.022 (0.005)	0.000 (0.000)	0.023 (0.005)	0.023 (0.005)	0.000 (0.000)	0.023 (0.005)	0.023 (0.005)	0.000 (0.000)	0.022 (0.005)	0.023 (0.005)
Per capita personal income	0.306 (0.051)	0.319 (0.081)	0.306 (0.051)	0.309 (0.077)	0.311 (0.080)	0.306 (0.051)	0.309 (0.077)	0.311 (0.081)	0.305 (0.051)	0.308 (0.077)	0.311 (0.080)
Establishment count	3629	85,949	3579	3579	15,894	3540	3540	15,429	3422	3422	14,923
Panel B : Your Economy Time Series (YTS)											
Number of employees	15,923 (66,360)	18,125 (24,849)	16,033 (66,942)	15,983 (21,505)	16,552 (22,089)	15,924 (67,800)	16,029 (21,630)	16,579 (22,291)	15,981 (69,319)	15,918 (21,752)	16,554 (22,679)
Median home price	0.174 (0.054)	0.121 (0.045)	0.174 (0.054)	0.126 (0.043)	0.127 (0.045)	0.174 (0.054)	0.126 (0.043)	0.127 (0.044)	0.173 (0.054)	0.125 (0.043)	0.126 (0.044)
Gasoline price	1.561 (0.508)	1.409 (0.494)	1.560 (0.508)	1.412 (0.490)	1.415 (0.490)	1.560 (0.508)	1.413 (0.490)	1.415 (0.491)	1.560 (0.508)	1.413 (0.490)	1.416 (0.490)

(Continues)

TABLE 1 (Continued)

Variable	Level of variation	(1) Treated unmatched	(2) Comparison unmatched	(3) Treated wide caliper	(4) Comparison max 1 NN	(5) Comparison max 5 NN	(6) Treated mid caliper	(7) Comparison max 1 NN	(8) Comparison max 5 NN	(9) Treated narrow caliper	(10) Comparison max 1 NN	(11) Comparison max 5 NN
Effective total tax rate	State	0.058 (0.004)	0.056 (0.010)	0.058 (0.004)	0.059 (0.010)	0.059 (0.010)	0.058 (0.004)	0.059 (0.010)	0.059 (0.010)	0.058 (0.004)	0.059 (0.010)	0.059 (0.010)
Effective income tax rate	State	0.041 (0.003)	0.014 (0.012)	0.041 (0.003)	0.017 (0.011)	0.017 (0.011)	0.041 (0.003)	0.017 (0.011)	0.017 (0.011)	0.041 (0.003)	0.017 (0.011)	0.017 (0.011)
Effective sales tax rate	State	0.000 (0.000)	0.022 (0.005)	0.000 (0.000)	0.022 (0.005)	0.022 (0.005)	0.000 (0.000)	0.022 (0.005)	0.022 (0.005)	0.000 (0.000)	0.022 (0.005)	0.022 (0.005)
Per capita personal income	County	0.306 (0.051)	0.318 (0.080)	0.306 (0.051)	0.312 (0.078)	0.313 (0.080)	0.306 (0.051)	0.311 (0.077)	0.312 (0.079)	0.306 (0.051)	0.310 (0.077)	0.311 (0.079)
Establishment count		3478	86,394	3414	3414	15,127	3327	3327	14,507	3172	3172	13,626

Note: The full sample period is from 2000 to 2007. Descriptive statistics for the matched samples are from the propensity score equation that predicts treatment based on the restaurant type (four categories), median household income in the census block in 2000, and the average annual growth rate in the census block group from 1990 to 2000 in population, poverty rate, median home value, and number of housing units. Means for each group and matched sample are shown. Standard deviations in parentheses. Unmatched establishment counts are the number of restaurants (N = 7225) located in non-border counties that are observed at least once in both the pre- (2000–2002) and post-treatment periods (2003–2007). Max 1 and 5 NN denote the 1- k nearest neighbor matches, up to a maximum of k , for a treated unit based on the closeness of the propensity score among all comparison units whose propensity scores fall within the caliper range (wide—mid—narrow) of the treated establishment. Comparison units are matched to treated units without replacement, and treated units are matched sequentially starting with the unit having the highest propensity score (which is the most difficult to match). Median home prices (county level), in millions, are from the Federal Housing and Finance Association. Per capita personal income (county level), in hundreds of thousands of dollars, is from the Bureau of Economic Analysis. State-level retail gasoline prices, in dollars per gallon, are from the U.S. Energy Information Administration. Effective state tax rates are the sum of state and local taxes collected, from the Census Bureau's Quarterly Summary of State & Local Tax Revenue, expressed as a fraction of personal income. Finally, census block group data for 1990 and 2000 used in the matching step are from IPUMS National Historical Geographic Information System (Manson et al., 2022).

establishment-level mean employment differs across treated and untreated groups by 19.3% in the NETS sample and 13.8% in the YTS sample.

In contrast, differences in mean employment levels across groups shrink substantially using the weighted (or matched) samples, despite the fact that employment counts were *never* used in the matching process. The weighted descriptive statistics are shown in columns (3) through (11). As an example, the mean employment among restaurants in Oregon is 15.46 using the wide caliper and the NETS sample (column 3 in Panel A). This figure differs from the unweighted mean (column 1) because some treated establishments go unmatched and are discarded. In this matched comparison group, mean employment counts for the single nearest neighbor matches (column 4 in Panel A) and 1:5 nearest neighbor matches (column 5) are 15.36 and 15.96, respectively. Across all three calipers, the absolute value of the difference between mean employment counts in Oregon restaurants and the comparison samples averages just 0.28 employees. This is a noticeable reduction from the unweighted (mean) difference of 2.96 workers. Panel B shows similar improvements in employment balance using the YTS sample. Thus, preprocessing the data to form matched samples based on the restaurant type, block group median household income in 2000, and average annual growth rates in several block group characteristics between 1990 and 2000 greatly improved the balance between treated and untreated establishments.

Varying the propensity score calipers and using variable ratio matching provide a simple robustness check because individual restaurants are matched to restaurants in different locations across the country (as long as they share the key characteristics). It is worthwhile to illustrate this for the NETS sample when matches are performed in sequential order matching using the widest caliper. Those descriptive statistics are shown in columns (3), (4), and (5) of Panel A in Table 1.

Out of a total of 3629 (non-border) county restaurants in Oregon, a total of 3579 (or 98.6%) were matched to one or more comparison restaurants. Of the 50 restaurants that were unmatched, 36 are located in Marion County, 5 are in Lane County, and 2 are located in Washington, Union, and Clackamas Counties, respectively. The remaining unmatched restaurants are from Linn County (1), Deschutes County (1), and Benton County (1). In percentage terms, unmatched restaurants only account for around 1% of the total number of restaurants, but this percentage is higher in Marion County (7.8%) and Union County (3.5%).

In terms of the matched establishments in this example, we illustrate spatial intensity and differences using single-nearest-neighbor and variable ratio matching in Figure 4. The top panel shows counts of the number of restaurants in each county matched to an Oregon restaurant using single-nearest-neighbor matching, while the bottom panel shows counts using variable ratio matching. There are a total of 1121 (non-border) counties in the 21 comparison states.

Using single-nearest-neighbor matching (top panel), in which each of the 3579 matched establishments in Oregon is paired to a single establishment in a comparison state, at least one restaurant from 644 counties was matched to an establishment in Oregon. This represents 57% of the comparison counties and illustrates the degree of geographic dispersion. The top locations for matches, shown in dark blue in the figure, included Salt Lake County, UT (205 matches), Harris County, TX (144 matches), and Travis County, TX (111 matches). In percentage terms, restaurants in those counties accounted for 5.7%, 4.0%, and 3.1% of the total matched establishments, respectively. In the counties with matched restaurants, 578 have fewer than 10 matches and the median number of matches per county is 2.

As noted earlier, the variable ratio matching attempts to match each establishment once before searching for higher-order matches. Using the wide caliper and NETS sample, we are able (again) to match 3579 restaurants in Oregon to a total of 15,894 restaurants located in 933 counties (bottom panel in Figure 4). This expands the number of counties with matches from 57% of the pool of 1121 counties to more than 83%. Each restaurant in Oregon is matched to an average of 4.4 comparison restaurants, but this varies from a low of 1 match to a maximum of 5.

There are also some differences in the locations where matches are selected using the variable ratio matching. While Salt Lake County, UT, and Harris County, TX remain the top two locations for matches, a sizable number of matches are also now found in Fulton County, GA (475 matches), Dallas County, TX (311), Marion County, IN (260), Davidson County, TN (225), and Jefferson Parish, LA (204). The 1:k matches are also less spatially concentrated, with the top five counties accounting for 15% of all matches, compared to 21% using single nearest neighbor matching. Of the 933 counties with matched establishments in the 1:k process, the median number of matches per county is 7.

Across all datasets, samples, and calipers (a total of 48 regressions), the mean and median percentage of Oregon's restaurants that are successfully matched is 97.8% and 98.4%, respectively. The smallest share matched, 94.6%, occurs using the YTS sample when non-employer establishments are omitted and we use the narrow caliper. At the other extreme, matches are found for 99.3% of Oregon's restaurants using the NETS sample with the widest caliper when non-employer establishments are omitted.

County Intensity of Restaurant Matches

Wide Caliper, NETS Database, Sequential Matching

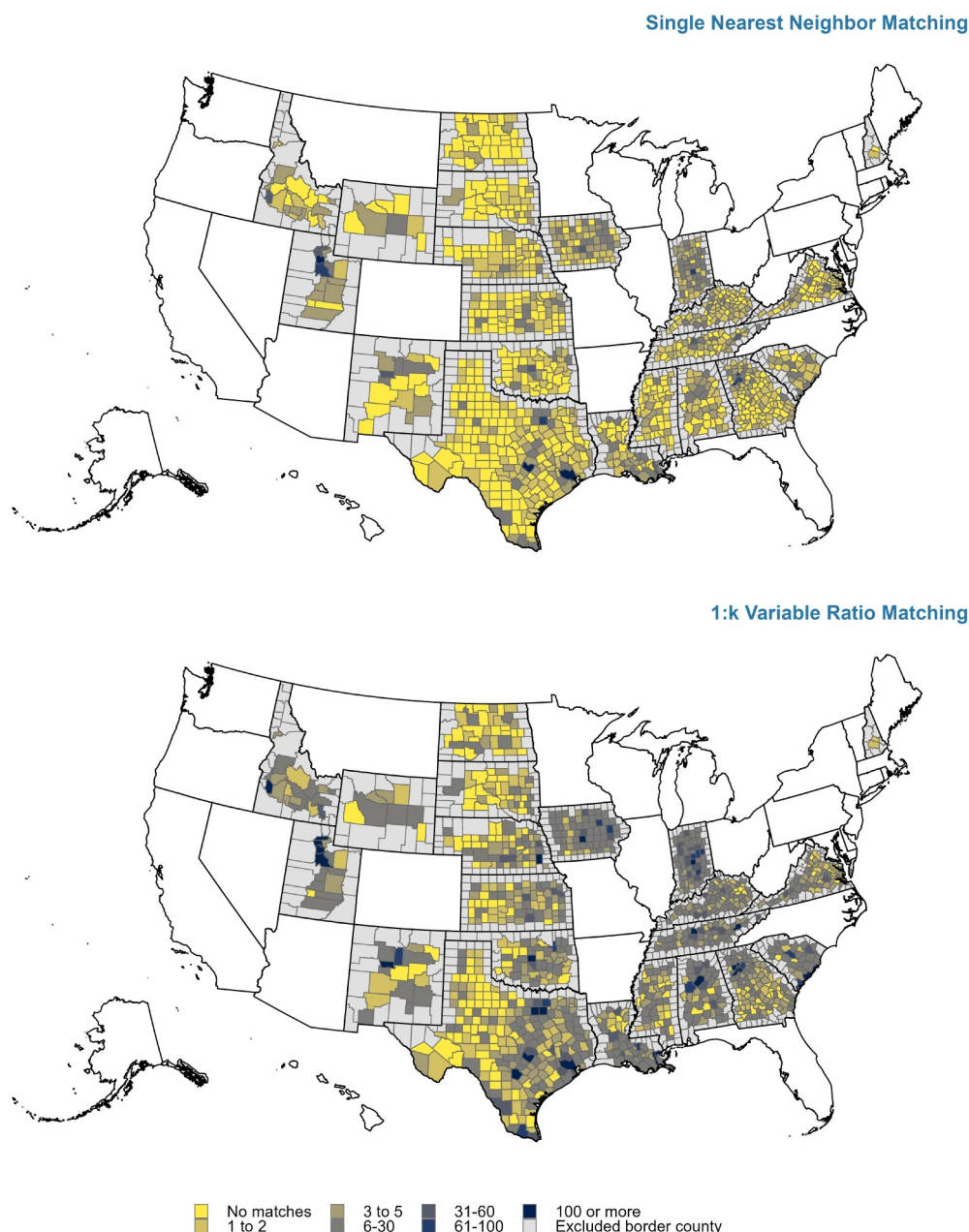


FIGURE 4 Location of restaurant matches. This figure shows the raw counts of restaurants in a given county that are matched to a restaurant in Oregon. Results are shown using the wide caliper and National Establishment Time Series (NETS) database. The top and bottom figures correspond to the samples shown in columns 4 and 5 of Panel 1 in Table 1.

It is worthwhile to briefly analyze unmatched restaurants under the baseline scenario that uses sequential order matching (hardest to match first) and the widest caliper. With these parameters, matches are found for 98.6% of the NETS sample and 98.2% of the YTS samples. In terms of counts, 50 restaurants are unmatched in the NETS sample and 64 are unmatched in the YTS sample. The majority of these unmatched restaurants, 40 in the NETS sample and 33 in the YTS sample, are located in the same census block group that encompasses some of the downtown riverfront area in Salem (Marion County).¹⁷ Among the 30,260 census block groups in our datasets, this area ranked in the bottom fifth percentile in population and in the 88th percentile in poverty rate in 1990. Over the next decade, however, revitalization took hold and this block group's population growth and median home value over the next decade ranked in the 88th

and 99th percentiles, respectively.¹⁸ There are no other block groups in the dataset that experienced a similar transformation, which explains why these restaurants are unmatched.

More generally, while the match percentages are fairly high across the set of regressions, it is important to keep in mind that the spatial location of matched restaurants differs using different datasets, samples, and calipers, sometimes considerably. In our particular setting, we view this as advantageous because of the natural robustness check it provides.

4 RESULTS

Baseline post-matching regression results are presented in Table 2. Estimation is by weighted-least squares using weights obtained from the matching process.¹⁹ Since the dependent variable is the log of establishment employment, the mean percentage change in establishment-level employment after indexing is given by $100 * (e^{\delta} - 1)$, where δ is the parameter of interest (Giles, 2011). For small parameter estimates, such as those in this study, the percentage change is closely approximated with $100 * \delta$.

Matched samples from the NETS database are presented in columns (1) through (6). Single nearest neighbor and variable ratio matched samples are shown side-by-side for the wide caliper (columns 1–2), mid caliper (columns 3–4), and narrow caliper (columns 5–6). These specifications are repeated in columns (7) through (12) when matched samples are formed from the YTS database. For each specification, we also report the number of treated establishments, the number that are matched, and the number of comparison establishments in the weighted comparison group.

Regressions include establishment and year fixed effects that are not reported. Recent work by badie and Spiess (2022) shows that standard errors are generally invalid in post-matching regressions when the matching step is ignored. When matching is done without replacement, which is the process we follow, badie and Spiess (2022) shows that bootstrapped standard errors can be used for robust inference if the resampling is done on matched sets. All standard errors (and confidence intervals) are from 1000 bootstrap replications following badie and Spiess (2022).

As one might expect, there are differences in the matched samples for both datasets when the caliper changes. Using the NETS database, the number of Oregon restaurants that are matched (out of 3629) decreases from a high of 3579 using the wide caliper to a low of 3422 using the narrow caliper. In the YTS database, there is a slightly more pronounced reduction in matches as the caliper tightens, ranging from a high of 3414 matches (out of 3478) using the wide caliper to a low of 3172 using the narrow caliper.

Across all 12 specifications in Table 2, the results are robust. We find a significant reduction in establishment-level employment (at the 5% level or better) in Oregon restaurants post-indexing relative to the matched comparison groups. In terms of magnitude, the reductions in employment range from 3.1% to 4.4% in the NETS samples and 3.6%–4.5% in the YTS samples. Across all specifications, the median reduction is 3.9%. It is also worth repeating that restaurants in Oregon are being compared to different sets of comparison establishments because of variable ratio matching, different calipers, and different sources of establishment-level data.

Event study versions of the regressions using the NETS samples presented in columns (1) through (6) of Table 2 are shown in Figure 5. Plots using the YTS samples, reported in columns (7) through (12) of Table 2, are shown in Figure 6. The year before indexing, 2002, was normalized to zero in the event study plots.

There do not appear to be any meaningful pre-trend differences in employment between the treated and comparison establishments in Figures 5 and 6. One can also see a noticeable (and statistically significant) reduction in establishment employment in Oregon's restaurants in 2003 when indexing became effective.²⁰

4.1 Robustness checks

In this section, we compare the robustness of the baseline results to three alternative samples and matching procedures. First, we match Oregon's restaurants in random order rather than following Rosenbaum (2020) and matching restaurants starting with the highest propensity score, the most difficult treated unit to match. This means that once a restaurant from Oregon is randomly selected, it is either matched to its nearest neighbor among all of the matches within the caliper or discarded because no suitable comparisons exist. Identical to the baseline results, first-order matches are attempted for every (randomly selected) restaurant in 1:k variable ratio matching before higher-order matches for a given restaurant are explored. Matching in random order increases the likelihood that an Oregon

T B L E 2 Effect of indexing the minimum wage on restaurant establishment employment.

	<i>dependent variable</i> log (number of employees)											
	(1) WLS	(2) WLS	(3) WLS	(4) WLS	(5) WLS	(6) WLS	(7) WLS	(8) WLS	(9) WLS	(10) WLS	(11) WLS	(12) WLS
<i>D</i>	−0.031*** (0.011)	−0.040*** (0.011)	−0.032*** (0.011)	−0.040*** (0.010)	−0.035*** (0.012)	−0.044*** (0.011)	−0.044*** (0.011)	−0.037*** (0.010)	−0.044*** (0.010)	−0.036*** (0.010)	−0.045*** (0.011)	−0.035*** (0.011)
Median home price	0.231 (0.154)	0.386*** (0.150)	0.256 (0.164)	0.393** (0.156)	0.288* (0.163)	0.431*** (0.160)	0.343*** (0.143)	0.283* (0.151)	0.346** (0.147)	0.272* (0.156)	0.387*** (0.152)	0.287* (0.165)
Gasoline price	0.183** (0.078)	0.139* (0.074)	0.180** (0.079)	0.134* (0.073)	0.190** (0.082)	0.139* (0.076)	0.175** (0.077)	0.118 (0.076)	0.187** (0.078)	0.123 (0.075)	0.185** (0.080)	0.119 (0.076)
Effective total tax rate	0.627 (2.144)	−0.209 (1.663)	0.447 (2.126)	−0.297 (1.657)	0.517 (2.149)	−0.133 (1.632)	−1.002 (2.043)	−1.324 (1.729)	−1.366 (2.007)	−1.656 (1.680)	−1.986 (2.087)	−2.004 (1.819)
Effective income tax rate	0.106 (2.483)	1.143 (1.905)	0.280 (2.455)	1.289 (1.897)	0.347 (2.498)	1.330 (1.927)	0.521 (2.375)	0.730 (1.998)	0.898 (2.381)	0.830 (1.954)	1.690 (2.412)	1.498 (2.044)
Effective sales tax rate	−3.626 (3.317)	−2.050 (2.637)	−3.733 (3.400)	−2.267 (2.657)	−2.998 (3.401)	−1.622 (2.660)	6.436** (3.093)	5.050** (2.449)	6.852** (3.147)	5.265** (2.504)	7.784** (3.095)	6.191** (2.481)
Per capita personal income	−0.211 (0.184)	−0.210 (0.141)	−0.178 (0.178)	−0.189 (0.137)	−0.208 (0.187)	−0.260* (0.145)	0.097 (0.216)	−0.125 (0.199)	0.153 (0.220)	−0.048 (0.202)	0.114 (0.223)	−0.037 (0.215)
<i>N</i>	51,358	140,176	50,789	136,598	49,092	132,152	45,600	123,975	44,423	119,290	42,382	112,332
Establishment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Treated estabs	3629	3629	3629	3629	3629	3629	3478	3478	3478	3478	3478	3478
Treated matches	3579	3579	3540	3540	3422	3422	3414	3414	3327	3327	3172	3172
Comparison estabs	85,949	85,949	85,949	85,949	85,949	85,949	86,394	86,394	86,394	86,394	86,394	86,394
Comparison matches	3579	15,894	3540	15,429	3422	14,923	3414	15,127	3327	14,507	3172	13,626
Counties in comp group	644	933	644	930	643	926	577	894	574	889	564	879
States in comp group	21	21	21	21	21	21	21	21	21	21	21	21
PScore caliper	Wide	Wide	Mid	Mid	Narrow	Narrow	Wide	Wide	Mid	Mid	Narrow	Narrow
Dataset	NETS	NETS	NETS	NETS	NETS	NETS	YTS	YTS	YTS	YTS	YTS	YTS
Max neighbor matches	1	5	1	5	1	5	1	5	1	5	1	5

Note: This table presents matched sample difference-in-differences estimates of the effect of Oregon indexing the minimum wage to the consumer price index in 2003 on individual restaurant establishment level employment. *D* is the difference-in-differences parameter of interest. Standard errors are from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022).

bbreviations: FE, fixed effects; NETS, National Establishment Time Series; WLS, weighted least squares; YTS, Your Economy Time Series.

***Denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, establishment fixed effects, and year fixed effects that are not reported. Columns 1 through 6 use matches from the propensity score equation and the NETS database. Columns 7 through 12 use matches from the propensity score equation and the YTS database.

Sequential Matched Samples: NETS

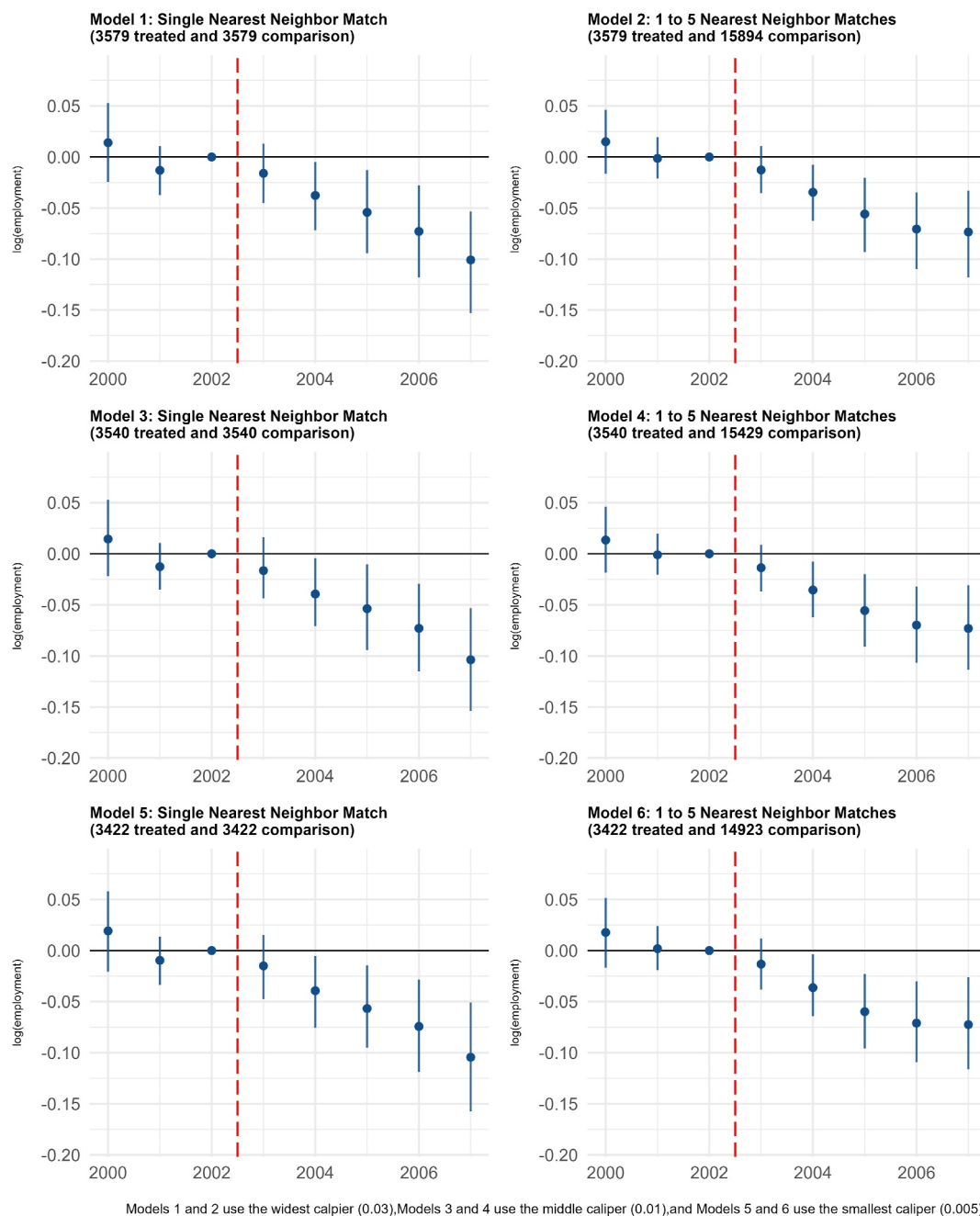


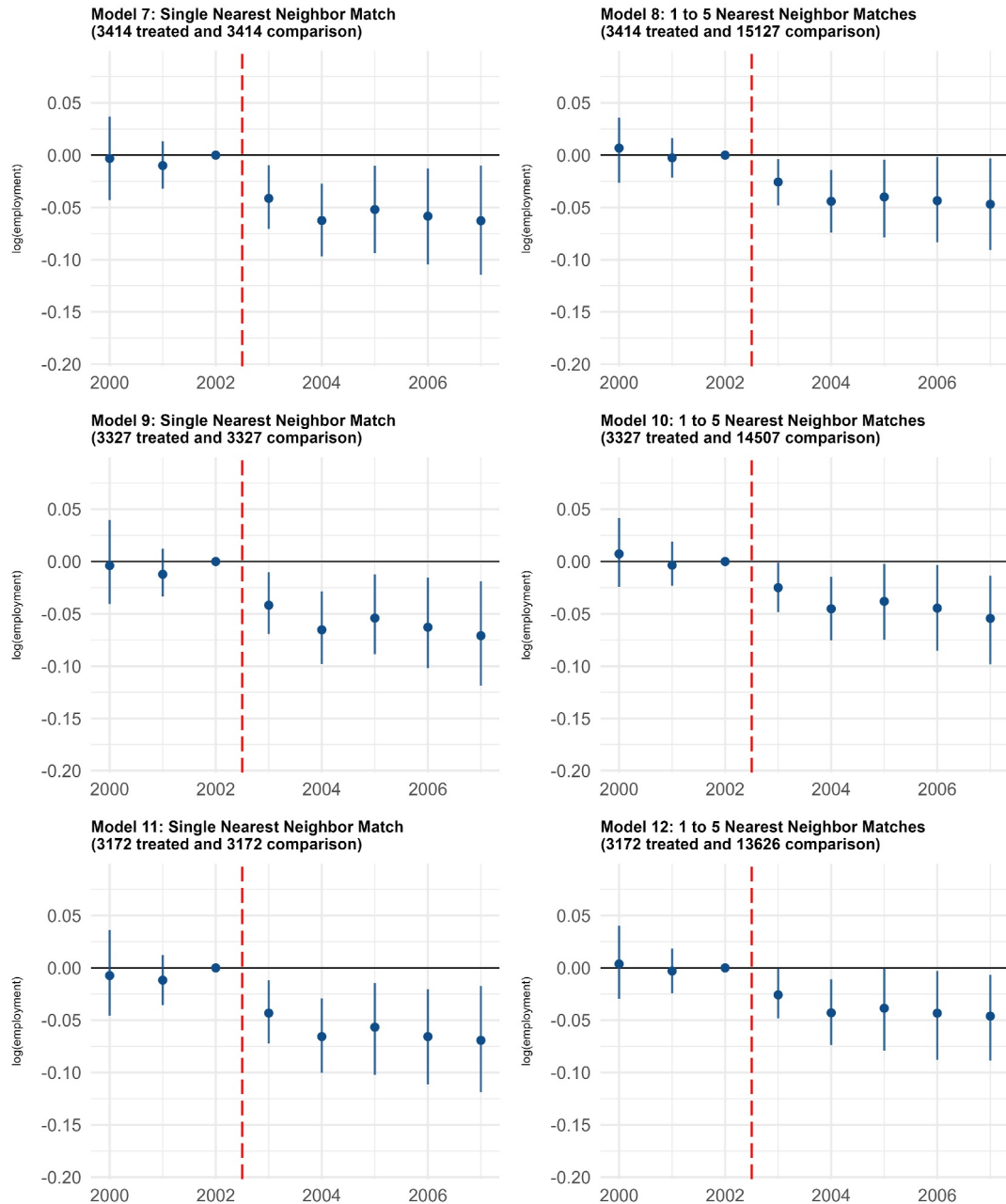
FIGURE 5 Event study plots: sequential matching using NETS database. Each panel shows the event study version of the matched sample regressions presented in Table 2. Model numbers in the figure correspond to the model numbers presented in Table 2. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). NETS, National Establishment Time Series.

establishment will be matched with one (or more) establishments that differ from those in the comparison groups that were formed using sequential order matching.

The remaining robustness checks exclude non-employer and small establishments. While one could argue that including non-employer establishments may be important because a higher minimum wage may lower the incentive to add a paid employee, this employment concept differs from official government sources, so omitting these establishments should shrink a key source of discrepancy between NETS, YTS, and official sources.

To accomplish this, we first omit all establishments from the sample that always report a single employee. As shown in Panel a of Figure 3, this is 2.8% and 2.4% of the NETS and YTS samples, respectively. Propensity scores are re-

Sequential Matched Samples: YTS



Models 1 and 2 use the widest caliper (0.03), Models 3 and 4 use the middle caliper (0.01), and Models 5 and 6 use the smallest caliper (0.005).

FIGURE 6 Event study plots: sequential matching using YTS database. Each panel shows the event study version of the matched sample regressions presented in Table 2. Model numbers in the figure correspond to the model numbers presented in Table 2. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). YTS, Your Economy Time Series database.

estimated for the adjusted NETS and YTS samples, and new weighted comparison groups are formed. We then expand the definition of small establishments to those always reporting two (or fewer) employees and repeat the same process. The NETS and YTS samples shrink by 6.5% and 7.7%, respectively, when these “small” establishments are excluded.²¹

Estimates matching in random order are presented in Table 3, and results excluding non-employer establishments and small establishments are presented in Tables 4 and 5, respectively. The presentation of results is identical to the baseline results in Table 2. Columns (1) through (6) use the NETS sample and various calipers, while columns (7) through (12) use the YTS sample.

T B L E 3 Random order matching.

	<i>ependent variable</i> log (number of employees)											
	(1) WLS	(2) WLS	(3) WLS	(4) WLS	(5) WLS	(6) WLS	(7) WLS	(8) WLS	(9) WLS	(10) WLS	(11) WLS	(12) WLS
<i>D</i>	−0.030** (0.012)	−0.040*** (0.011)	−0.032*** (0.012)	−0.037*** (0.011)	−0.036*** (0.012)	−0.039*** (0.011)	−0.041*** (0.011)	−0.034*** (0.010)	−0.042*** (0.010)	−0.032*** (0.010)	−0.043*** (0.011)	−0.032*** (0.011)
Median home price	0.200 (0.156)	0.387** (0.153)	0.207 (0.157)	0.342** (0.150)	0.256 (0.167)	0.373** (0.159)	0.305** (0.151)	0.216 (0.157)	0.299** (0.150)	0.214 (0.154)	0.377** (0.156)	0.229 (0.154)
Gasoline price	0.175** (0.078)	0.130* (0.074)	0.179** (0.082)	0.117 (0.076)	0.191** (0.084)	0.133* (0.076)	0.171** (0.079)	0.118 (0.074)	0.179** (0.076)	0.119 (0.076)	0.176** (0.081)	0.120 (0.074)
Effective total tax rate	0.786 (2.160)	0.208 (1.723)	0.583 (2.132)	0.069 (1.672)	0.785 (2.122)	0.051 (1.701)	−0.642 (2.111)	−1.088 (1.768)	−0.923 (2.162)	−1.035 (1.771)	−1.608 (2.132)	−1.448 (1.802)
Effective income tax rate	−0.188 (2.454)	0.414 (1.941)	0.002 (2.459)	0.733 (1.948)	−0.317 (2.481)	0.865 (1.976)	0.252 (2.505)	0.634 (1.993)	0.501 (2.581)	0.274 (2.027)	1.161 (2.529)	0.905 (2.040)
Effective sales tax rate	−3.592 (3.351)	−2.316 (2.680)	−3.377 (3.359)	−2.425 (2.693)	−3.877 (3.368)	−2.295 (2.628)	5.335* (3.173)	3.362 (2.521)	5.648* (3.270)	3.892 (2.633)	5.948* (3.363)	4.379* (2.528)
Per capita personal income	−0.239 (0.186)	−0.288** (0.133)	−0.201 (0.192)	−0.251* (0.148)	−0.204 (0.188)	−0.272* (0.144)	0.114 (0.218)	−0.111 (0.200)	0.153 (0.214)	−0.053 (0.204)	0.141 (0.234)	−0.047 (0.219)
<i>N</i>	51,323	140,154	50,668	136,472	49,169	132,084	45,466	123,764	44,314	119,126	42,498	112,437
Establishment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Treated estabs	3629	3629	3629	3629	3629	3629	3478	3478	3478	3478	3478	3478
Treated matches	3579	3579	3534	3533	3428	3421	3406	3409	3318	3318	3177	3178
Comparison estabs	85,949	85,949	85,949	85,949	85,949	85,949	86,394	86,394	86,394	86,394	86,394	86,394
Comparison matches	3579	15,893	3534	15,421	3428	14,916	3406	15,105	3318	14,494	3177	13,629
Counties in comp group	643	935	644	931	642	927	578	893	576	889	567	879
States in comp group	21	21	21	21	21	21	21	21	21	21	21	21
PScore caliper	Wide	Wide	Mid	Mid	Narrow	Narrow	Wide	Wide	Mid	Mid	Narrow	Narrow
Dataset	NETS	NETS	NETS	NETS	NETS	NETS	YTS	YTS	YTS	YTS	YTS	YTS
Max neighbor matches	1	5	1	5	1	5	1	5	1	5	1	5

Note: This table presents matched sample difference-in-differences estimates of the effect of Oregon indexing the minimum wage to the consumer price index in 2003 on individual restaurant establishment level employment. *D* is the difference-in-differences parameter of interest. Standard errors are from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022).

bbreviations: FE, fixed effects; NETS, National Establishment Time Series; WLS, weighted least squares; YTS, Your Economy Time Series.

***Denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, establishment fixed effects, and year fixed effects that are not reported. Columns 1 through 6 use matches from the propensity score equation and the NETS database. Columns 7 through 12 use matches from the propensity score equation and the YTS database.

T B L E 4 Non-employer establishments excluded.

	<i>dependent variable</i> log (number of employees)											
	(1) WLS	(2) WLS	(3) WLS	(4) WLS	(5) WLS	(6) WLS	(7) WLS	(8) WLS	(9) WLS	(10) WLS	(11) WLS	(12) WLS
<i>D</i>	−0.041*** (0.012)	−0.041*** (0.011)	−0.040*** (0.013)	−0.039*** (0.012)	−0.043*** (0.013)	−0.042*** (0.012)	−0.028*** (0.010)	−0.028*** (0.010)	−0.031*** (0.010)	−0.031*** (0.011)	−0.026*** (0.011)	−0.026*** (0.011)
Median home price	0.346** (0.167)	0.301* (0.177)	0.329** (0.164)	0.280 (0.174)	0.367** (0.159)	0.281 (0.171)	0.169 (0.147)	0.204 (0.163)	0.222 (0.142)	0.275* (0.161)	0.207 (0.146)	0.259 (0.158)
Gasoline price	0.139* (0.083)	0.217*** (0.079)	0.126 (0.084)	0.197** (0.079)	0.154* (0.079)	0.232*** (0.075)	0.114 (0.071)	0.141** (0.069)	0.110 (0.077)	0.128* (0.073)	0.106 (0.077)	0.097 (0.078)
Effective total tax rate	5.594*** (2.129)	4.743*** (1.816)	5.339** (2.169)	4.641*** (1.791)	5.073** (2.080)	4.304** (1.693)	2.790 (2.218)	1.253 (1.949)	3.331 (2.335)	1.817 (2.030)	3.194 (2.431)	2.113 (2.173)
Effective income tax rate	−6.643*** (2.508)	−5.319** (2.105)	−6.270** (2.516)	−5.218** (2.053)	−5.921** (2.447)	−5.022** (1.991)	−2.870 (2.479)	−0.802 (2.162)	−3.654 (2.624)	−1.598 (2.184)	−3.576 (2.714)	−2.035 (2.379)
Effective sales tax rate	−7.973** (3.325)	−7.403*** (2.760)	−7.261** (3.428)	−6.645** (2.846)	−6.792* (3.507)	−6.116** (2.871)	−2.874 (3.232)	−1.813 (2.788)	−3.554 (3.430)	−2.417 (2.928)	−3.312 (3.471)	−2.556 (3.013)
Per capita personal income	−0.281* (0.167)	−0.400*** (0.145)	−0.259 (0.171)	−0.372** (0.145)	−0.258 (0.168)	−0.348** (0.147)	−0.017 (0.170)	−0.132 (0.140)	0.016 (0.172)	−0.073 (0.149)	0.029 (0.173)	−0.045 (0.147)
<i>N</i>	50,441	137,223	49,371	132,470	48,029	127,652	44,740	121,895	43,115	118,320	41,239	110,623
Establishment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Treated estabs	3629	3629	3629	3629	3629	3629	3478	3478	3478	3478	3478	3478
Treated matches	3502	3502	3428	3428	3331	3331	3338	3338	3217	3217	3077	3077
Comparison estabs	85,949	85,949	85,949	85,949	85,949	85,949	86,394	86,394	86,394	86,394	86,394	86,394
Comparison matches	3502	15,492	3428	14,906	3331	14,329	3338	14,764	3217	14,357	3077	13,361
Counties in comp group	617	938	614	936	609	933	577	875	576	873	574	863
States in comp group	21	21	21	21	21	21	21	21	21	21	21	21
PScore caliper	Wide	Wide	Mid	Mid	Narrow	Narrow	Wide	Wide	Mid	Mid	Narrow	Narrow
Dataset	NETS	NETS	NETS	NETS	NETS	NETS	YTS	YTS	YTS	YTS	YTS	YTS
Max neighbor matches	1	5	1	5	1	5	1	5	1	5	1	5

Note: This table presents matched sample difference-in-differences estimates of the effect of Oregon indexing the minimum wage to the consumer price index in 2003 on individual restaurant establishment level employment. *D* is the difference-in-differences parameter of interest. Standard errors are from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022).

bbreviations: FE, fixed effects; NETS, National Establishment Time Series; WLS, weighted least squares; YTS, Your Economy Time Series.

***Denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, establishment fixed effects, and year fixed effects that are not reported. Columns 1 through 6 use matches from the propensity score equation and the NETS database. Columns 7 through 12 use matches from the propensity score equation and the YTS database.

T B L E 5 Small establishments excluded.

	<i>dependent variable</i> log (number of employees)											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	WLS	WLS	WLS	WLS	WLS	WLS	WLS	WLS	WLS	WLS	WLS	WLS
<i>D</i>	−0.043*** (0.012)	−0.036*** (0.012)	−0.043*** (0.012)	−0.036*** (0.012)	−0.042*** (0.012)	−0.035*** (0.012)	−0.033*** (0.010)	−0.032*** (0.009)	−0.031*** (0.010)	−0.032*** (0.009)	−0.035*** (0.011)	−0.034*** (0.009)
Median home price	0.109 (0.160)	0.144 (0.158)	0.154 (0.154)	0.179 (0.159)	0.143 (0.163)	0.156 (0.165)	0.255* (0.135)	0.230* (0.133)	0.298** (0.138)	0.306** (0.134)	0.326** (0.136)	0.317** (0.130)
Gasoline price	0.208** (0.086)	0.150** (0.076)	0.187** (0.084)	0.124* (0.074)	0.198** (0.087)	0.117 (0.078)	0.075 (0.074)	0.044 (0.070)	0.060 (0.076)	0.033 (0.066)	0.065 (0.075)	0.022 (0.068)
Effective total tax rate	6.791*** (2.424)	2.036 (1.887)	6.696*** (2.426)	1.859 (2.009)	5.602** (2.394)	0.793 (1.921)	2.110 (2.545)	1.585 (2.121)	2.123 (2.768)	1.241 (2.178)	2.859 (2.603)	2.093 (2.085)
Effective income tax rate	−7.051** (2.899)	−1.980 (2.358)	−6.839** (2.881)	−1.777 (2.413)	−5.363* (2.856)	−0.442 (2.399)	−3.565 (2.852)	−1.928 (2.227)	−3.812 (3.106)	−1.802 (2.339)	−4.623 (2.942)	−2.555 (2.310)
Effective sales tax rate	−7.959** (3.703)	−5.065* (2.910)	−8.545** (3.613)	−5.528* (3.008)	−6.440* (3.745)	−2.977 (2.993)	−3.787 (3.550)	−1.371 (2.902)	−3.865 (3.767)	−0.835 (2.861)	−4.971 (3.835)	−1.975 (2.910)
Per capita personal income	−0.222 (0.195)	−0.321** (0.151)	−0.165 (0.192)	−0.247 (0.154)	−0.190 (0.205)	−0.256 (0.157)	−0.341** (0.148)	−0.341*** (0.122)	−0.388** (0.159)	−0.437*** (0.129)	−0.430*** (0.166)	−0.466*** (0.130)
<i>N</i>	48,400	132,364	47,064	127,500	45,705	123,455	42,901	116,213	41,841	111,643	39,377	105,403
Establishment FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Treated estabs	3629	3629	3629	3629	3629	3629	3478	3478	3478	3478	3478	3478
Treated matches	3351	3351	3258	3258	3165	3165	3176	3176	3101	3101	2917	2917
Comparison estabs	85,949	85,949	85,949	85,949	85,949	85,949	86,394	86,394	86,394	86,394	86,394	86,394
Comparison matches	3351	14,871	3258	14,286	3165	13,832	3176	14,023	3101	13,435	2917	12,699
Counties in comp group	644	927	642	923	640	915	515	861	513	851	509	848
States in comp group	21	21	21	21	21	21	20	21	20	21	20	21
PScore caliper	Wide	Wide	Mid	Mid	Narrow	Narrow	Wide	Wide	Mid	Mid	Narrow	Narrow
Dataset	NETS	NETS	NETS	NETS	NETS	NETS	YTS	YTS	YTS	YTS	YTS	YTS
Max neighbor matches	1	5	1	5	1	5	1	5	1	5	1	5

Note: This table presents matched sample difference-in-differences estimates of the effect of Oregon indexing the minimum wage to the consumer price index in 2003 on individual restaurant establishment level employment. *D* is the difference-in-differences parameter of interest. Standard errors are from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022).

bbreviations: FE, fixed effects; NETS, National Establishment Time Series; WLS, weighted least squares; YTS, Your Economy Time Series.

***Denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. Regressions were estimated with a constant term, establishment fixed effects, and year fixed effects that are not reported. Columns 1 through 6 use matches from the propensity score equation and the NETS database. Columns 7 through 12 use matches from the propensity score equation and the YTS database.

Whether we form matched samples in random order or exclude non-employer or other small establishments, we continue to find that establishment-level employment in Oregon restaurants fell after indexing relative to restaurants in these alternative comparison groups. The estimated magnitudes are quite similar to the baseline results as well. Forming matched samples in a random order (Table 3), the point estimates range from -3.0% to -4.2% , with a median estimate of -3.7% . When non-employer or other small establishments are omitted, which decreases the total sample size between 2.4% and 7.7% depending on the dataset and definition of small establishments, the median reduction is 3.5% in both scenarios.²² Finally, when we estimated an event study version for each of these additional regressions, there were no notable differences in pre-treatment employment levels. These additional figures are presented in Figures 1–6.

5 CONCLUSION

Indexing the minimum wage to inflation has become a popular way to address the disconnect between low-wage employees' purchasing power and high prices. In fact, 18 states will have implemented the policy by 2025. Many of these states will begin their inflation adjustments amid aggressive increases in the federal funds rate target and the pandemic that shocked employment in low-wage service occupations—two external factors likely to confound estimates of the effects of minimum wage changes. We provide a benchmark analysis for the employment effects of indexing minimum wage compared to an alternative of a constant nominal minimum wage rate.

Exploiting a period of relative macroeconomic stability, we study how indexing the minimum wage affected establishment-level restaurant employment in Oregon. Compared to a representative sample of establishments selected through matching methods, we find Oregon restaurants experienced lower employment—about 3.6% on average—after that state indexed its minimum wage to the Consumer Price Index in 2003. The findings are consistent across three alternative matching procedures. In the first, matching was conducted in random order instead of matching in order of the highest propensity score. Next, restaurants with a single employee during the entire sample period, or with two or fewer employees during the entire sample period, were sequentially omitted from the treated and comparison groups. These results were effectively identical to the baseline results, even using two distinct national business establishment datasets.

Before the policy change, Oregon had roughly 3600 (non-border county) restaurants with an average of 16 employees. Indexing increased the minimum wage by 20% between 2003 and 2007, resulting in more than 2000 lost jobs using our baseline median estimate. This implies an employment elasticity of -0.18 .

Unlike comparative case study or contiguous border designs that are prevalent in the minimum wage literature, our framework focuses on a single policy change and uses comparison establishments that do not share a common labor pool. Thus, it leverages key attributes of both sides of the minimum wage methodology debate.

Our implied employment elasticity from indexing is 60% smaller than the effect found by Brummund and Strain (2020), which is the most closely related study. Brummund and Strain (2020) compare county employment in states that indexed the minimum wage to states making periodic minimum wage adjustments. Our study examines how indexing affected non-border county establishment-level employment relative to matched non-border county establishments in states where the nominal minimum wage was unchanged.

With common labor pool and potential disemployment effects from establishment exit and entry removed from this study, our estimates provide a baseline of employment effect size when introducing an indexed minimum wage when the alternative is to maintain the current minimum wage policy. This has important policy implications for analyzing the success or failure of future indexing policies, and highlights the importance of considering long-term indexing policies in future minimum wage studies.

ACKNOWLEDGMENTS

We thank co-editor David Sims and two anonymous referees for several suggestions that have improved the manuscript. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Replication data and code are available at Open ICPSR project Miller et al. (2025). Any errors are our own.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Open ICPSR Project “ECIN Replication Package for Employment Effects of Minimum Wage Indexing: Establishment Evidence from Oregon Restaurants” at <http://doi.org/10.3886/E208522V4>.

ORCID

Gary A. Wagner  <https://orcid.org/0000-0002-7493-8419>
 Alicia Plemmons  <https://orcid.org/0000-0002-1573-4686>

ENDNOTES

- ¹ Source: Vaghul, Kavya and Zipperer, Ben (2016). “Historical State and Sub-State Minimum Wage Data,” Washington Center for Equitable Growth (<https://equitablegrowth.org/working-papers/historical-state-and-sub-state-minimum-wage-data/>).
- ² more detailed description of the Your Economy Time Series (YTS) database is available at the University of Wisconsin Business Dynamics Research Consortium’s website: <https://youreconomy.org/download/ytsdescription.pdf> (accessed: 11/12/2022).
- ³ The NETS database includes establishments with a single employee, so its coverage of small establishments is greater than what is provided by other federal sources of employment (Zeballos & Marchesi, 2022). Neumark et al. (2007) note that subtracting one from employer counts in NETS more closely approximates the conventional wage-and-salary employment concept using data for California. Due to the concern with small establishments and imputation, Zeballos and Marchesi (2022) redo their comparison with other official government data sources in the food-related industries after removing establishments with fewer than 2, 5, and 10 employees, respectively, from the NETS data. In doing so, they note that “for food-away-from-home trends, not much difference existed between NETS and official records when removing establishments of varying employment headcounts” (page 20). They also note that “in general, the national-level patterns hold across most States, suggesting there is little variation that is likely attributed to non-employer establishments in the food-away-from-home sector between NETS and iCBP” (page 20). Thus, while non-employer establishments reported in the NETS data may be an important consideration in some industries, Zeballos and Marchesi (2022) show that it is not a major concern in the (food-away-from-home) restaurant sector. In our NETS sample, fewer than 3% of establishments are reported as always having a single employee. As a robustness check, we exclude establishments that always reported a single employee, as well as those that always reported two or fewer employees, from the treated and comparison groups and redo our analysis. This did not lead to any meaningful changes in our empirical findings.
- ⁴ The University of Wisconsin’s Business Dynamics Research Consortium has created a summary comparing the Your Economy Time Series (YTS) database to several official data sources, including the QCEW and CBP. This information may be found here: <https://wisconsinbdr.org/comparing-economic-data-sources-2/> (accessed: 12/30/2023).
- ⁵ small number of the 4-digit NICS year-county observations in Oregon during the sample period were suppressed in County Business Patterns (CBP) and the Quarterly Census of Employment and Wages (QCEW). We use the imputed County Business Patterns, created by Eckert et al. (2020), for the employment measure from this data source. Suppressed QCEW cells (county by year by NICS) were imputed using the Kalman filter based on observed county by year by NICS quarterly data from 1990 to 2007.
- ⁶ Border counties were determined by the Census Bureau’s county adjacency file, available at <https://www.census.gov/geographies/reference-files/2010/geo/county-adjacency.html> (accessed: 5/5/2022).
- ⁷ It is, of course, an interesting question to explore how indexing may have affected entry and exit decisions, but this is beyond the scope of the current study.
- ⁸ Details on the imputed County Business Patterns (iCBP) are available in Eckert et al. (2020).
- ⁹ Some employment figures are imputed in the NETS and YTS databases. The NETS database flags actual versus estimated employment counts. In our sample, 71.3% of the observations are flagged as actual employment counts, with the remainder being estimates. In contrast, the YTS database does not flag estimated employment figures until 2020. According to the staff at Wisconsin’s Business Dynamics Research Consortium, roughly half of the YTS observations during our sample period are estimates.
- ¹⁰ Several of the better-known studies combining matching and regression include Rubin (1979), Heckman et al. (1998), and Rubin and Thomas (2000).
- ¹¹ Census block group data and shapefiles for the 1990 and 2000 Censuses were obtained from IPUMS NHGIS, University of Minnesota, www.nhgis.org. Restaurants were mapped to 2000 census block groups using the latitude and longitude fields reported in the NETS and YTS databases. A crosswalk mapping 1990 block groups to 2000 block groups was constructed using the percentage of the block group areas that overlapped based on the shapefiles.
- ¹² We also explored forming matched samples using Mahalanobis distance, which matches directly on covariate proximity rather than the probability of being assigned into the treatment group. Mahalanobis matching can perform poorly in the presence of indicator variables because it gives greater weight to variables with smaller variances (Gu & Rosenbaum, 1993). In our application, we include four indicator

variables in the matching process and Mahalanobis distance matching always led to inferior group balance when compared to propensity score matching.

- ¹³ In the variable ratio matching, we follow Rosenbaum (2020) and match treated units sequentially to their closest neighbor before additional higher-order matches are considered.
- ¹⁴ Imbens and Rubin (2015) discuss the trade-offs between internal and external validity in chapter 16.
- ¹⁵ Annual time-varying data at the block group or tract level are not readily available until the 5-year American Community Survey was released in 2009.
- ¹⁶ In the case of single nearest neighbor matching, all comparison units are paired to a single treated unit, so these units are assigned a weight of unity. In variable ratio matching, each comparison unit matched to a given treated unit is weighted by $p/(1 - p)$, where p is the proportion of units in the subclass assigned to the treated group.
- ¹⁷ The 2000 census block group GEOID for this area is 410,470,002,004.
- ¹⁸ The City of Salem created a riverfront-downtown renewal area in 1975 to revitalize this area. There have been several tax incentives and public-private investments in the region that could be partly responsible for the rapid population and home price growth observed between 1990 and 2000. The renewal plan, which includes historical information, is available here: <https://www.cityofsalem.net/home/showpublisheddocument/20755/638314292832200000> (accessed: 1/22/2024).
- ¹⁹ Matching weights are discussed in Footnote 16. Since comparison units are assigned a weight of unity in single-nearest-neighbor matching, weighted least squares (WLS) is equivalent to ordinary least squares in these specifications. This is not the case with variable ratio matching, so we refer to estimation as WLS to avoid unnecessary confusion.
- ²⁰ As Figure 2 illustrates, restaurant counts differ between NETS and YTS even though they have the same goal of capturing the universe of establishments. Because they are produced by different vendors, there is no precise way to match establishments across datasets. One would have to rely on longitude/latitude, establishment name, street address, or some combination of these attributes for matching. Differences in the point estimates will be an artifact of differences in the datasets. These factors could include: (i) different establishment counts, (ii) different employment counts for the same establishment, (iii) differences in the timing of employment changes, (iv) different imputation methods, or (v) a combination of these factors.
- ²¹ Across the full sample period, the National Establishment Time Series (NETS) database reports 98 restaurants in Oregon that always have a single employee and 240 restaurants that always have two (or fewer) employees. The comparable figures in the Your Economy Time Series database are 77 and 240, respectively.
- ²² As an additional robustness check, we re-estimated every specification and restricted the restaurant subclassification type (full-service, limited-service, etc) to be an exact match in the propensity score step. This reduced the percentage of treated restaurants in Oregon that were matched by approximately 6 percentage points. Across these 48 specifications, the median point estimate is -3.1% and the coefficients are significant at the 5% level or better in 42 regressions (87.5%). When combined with the results in the paper, we find a median point estimate of -3.3% that is significant at the 5% level or better in 89 out of 96 regressions.

REFERENCES

- badie, . (2005) Semiparametric difference-in-differences estimators. *The Review of Economic Studies*, 72(1), 1–19. available from: <https://doi.org/10.1111/0034-6527.00321>
- badie, . & Spiess, J. (2022) Robust post-matching inference. *Journal of the American Statistical Association*, 117(538), 983–995. available from: <https://doi.org/10.1080/01621459.2020.1840383>
- advisory Commission on Intergovernmental Relations. (1994) *Significant features of fiscal federalism*. Washington, DC: U.S. advisory Commission on Intergovernmental Relations.
- Barnatchez, K., Crane, L.D. & Decker, R. . (2017) An assessment of the National Establishment Time Series (NETS) database. Finance and Economics Discussion Series 2017–110, Board of Governors of the Federal Reserve System. available from: <https://doi.org/10.17016/feds.2017.110>
- Belsky, E. & Prakken, J. (2006). *Housing wealth effects: housing's impact on wealth accumulation, wealth distribution and consumer spending*. Joint Center for Housing Studies. Harvard University.
- Binder, C. & Makridis, C. (2022) Stuck in the seventies: gas prices and consumer sentiment. *The Review of Economics and Statistics*, 104(2), 293–305. ISSN: 0034-6535. available from: https://doi.org/10.1162/rest_a_00944
- Bivens, J. (2022) *Inflation, minimum wages, and profits: protecting low-wage workers from inflation means raising the minimum wage*. Economic Policy Institute Working Economics Blog (September 22).
- Blair, B.F. & Rezek, J.P. (2008) The effects of Hurricane Katrina on price pass-through for Gulf Coast gasoline. *Economics Letters*, 98(3), 229–234. ISSN: 0165-1765. available from: <https://doi.org/10.1016/j.econlet.2007.02.028>
- Brummund, P. & Strain, M.R. (2020) Does employment respond differently to minimum wage increases in the presence of inflation indexing? *Journal of Human Resources*, 55(3), 999–1024. available from: <https://doi.org/10.3368/jhr.55.2.1216.8404r2>

- Campello, M., Connolly, R., Kankanhalli, G. & Steiner, E. (2022) Do real estate values boost corporate borrowing? Evidence from contract-level data. *Journal of Financial Economics*, 144(2), 611–644. ISSN: 0304-405X. available from: <https://doi.org/10.1016/j.jfineco.2021.06.021>
- Card, D. (1992) Do minimum wages reduce employment? case study of California, 1987–1989. *Industrial and Labor Relations Review*, 46(1), 38–54. available from: <https://doi.org/10.2307/2524737>
- Card, D. & Krueger, A.B. (1994) Minimum wage and employment: a case study of the fast food industry in New Jersey and Pennsylvania. *The American Economic Review*, 84(4), 772–784.
- Card, D. & Krueger, A.B. (2000) Minimum wage and employment: a case study of the fast-food industry in New Jersey and Pennsylvania: reply. *The American Economic Review*, 90(5), 1397–1420. available from: <https://doi.org/10.1257/aer.90.5.1397>
- Clemens, J. (2021) How do firms respond to minimum wage increases? Understanding the relevance of non-employment margins. *The Journal of Economic Perspectives*, 35(1), 51–72. available from: <https://doi.org/10.1257/jep.35.1.51>
- Clemens, J. & Strain, M.B. (2017) *Estimating the employment effects of recent minimum wage changes: early evidence, an interpretative framework, and a pre-commitment to future analysis*. Working Paper 23084. National Bureau of Economic Research.
- Clemens, J. & Strain, M.B. (2021) *The heterogeneous effects of large and small minimum wage changes: evidence over the short and medium run using a pre-analysis plan*. Working Paper 29264. National Bureau of Economic Research.
- Clemens, J. & Strain, M.R. (2018) The short-run employment effects of recent minimum wage changes: evidence from the American Community Survey. *Contemporary Economic Policy*, 36(4), 711–722. available from: <https://doi.org/10.1111/coep.12279>
- Congressional Research Service. (2022) *State minimum wages: an overview*. R43792. available from: <https://sgp.fas.org/crs/misc/R43792.pdf>
- Courtemanche, C. (2011) Silver lining? The connection between gasoline prices and obesity. *Economic Inquiry*, 49(3), 935–957. available from: <https://doi.org/10.1111/j.1465-7295.2009.00266.x>
- Coyne, D. & Johnson, C. (2023) Use of the opportunity zone tax incentive: what the tax data tell us. Office of Tax Analysis, U.S. Department of the Treasury.
- Currie, J. & Fallick, B. (1996) The minimum wage and the employment of youth: evidence from the NLSY. *Journal of Human Resources*, 31(2), 401–428. available from: <https://doi.org/10.2307/146069>
- Dube, J., Lester, W. & Reich, M. (2010) Minimum wage effects across state borders: estimating using contiguous counties. *The Review of Economics and Statistics*, 92(4), 945–964. available from: https://doi.org/10.1162/rest_a_00039
- Eckert, F., Fort, T.C., Schott, P.R., & Yang, N.J. (2020) *Imputing missing values in the US Census Bureau's County Business Patterns*. Working Paper 26632. National Bureau of Economic Research. available from: <https://doi.org/10.3386/w26632>
- García, D.I. (2022) Second-home buying and the housing boom and bust. *Real Estate Economics*, 50(1), 33–58. available from: <https://doi.org/10.1111/1540-6229.12343>
- General Accountability Office. (2001) *Motor fuels: gasoline prices in Oregon*. Report G O-01-433R. General Accountability Office.
- Giles, D.E. (2011) Interpreting dummy variables in semi-logarithmic regression models: exact distributional results. *University of Victoria Department of Economics Working Paper EWP 1101*, pp. 1–24.
- Griffin, J.M., Kruger, S. & Maturana, G. (2021) What drove the 2003–2006 house price boom and subsequent collapse? Disentangling competing explanations. *Journal of Financial Economics*, 141(3), 1007–1035. ISSN: 0304-405X. available from: <https://doi.org/10.1016/j.jfineco.2020.06.014>
- Groizard, J.L., Ranjan, P. & Rodriguez-Lopez, J. (2015) Trade costs and job flows: evidence from establishment level data. *Economic Inquiry*, 53(1), 173–204. available from: <https://doi.org/10.1111/ecin.12139>
- Gu, X.S. & Rosenbaum, P.R. (1993) Comparison of multivariate matching methods: structures, distances, and algorithms. *Journal of Computational & Graphical Statistics*, 2(4), 405–420. available from: <https://doi.org/10.1080/10618600.1993.10474623>
- Heckman, J.J., Ichimura, H. & Todd, P. (1998) Matching as an econometric evaluation estimator. *The Review of Economic Studies*, 65(2), 261–294. available from: <https://doi.org/10.1111/1467-937X.00044>
- Ho, D.E., Imai, K., King, G. & Stuart, E.A. (2007) Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference. *Political Analysis*, 15(3), 199–236. available from: <https://doi.org/10.1093/pan/mpm013>
- Houde, J.-F., Newberry, P. & Seim, K. (2023) Nexus tax laws and economies of density in e-commerce: a study of Amazon's fulfillment center network. *Econometrica*, 91(1), 147–190. available from: https://doi.org/10.3982/ECT_15265
- Imbens, G.W. & Rubin, D.B. (2015) *Causal inference for statistics, social, and biomedical sciences: an introduction*. Cambridge University Press. available from: <https://doi.org/10.1017/CBO9781139025751>
- Jha, P., Neumark, D. & Rodriguez-Lopez, J. (2022) What's across the border? Re-evaluating the cross-border evidence on minimum wage effects. IZ Discussion Paper No. 15282.
- Katz, L. & Krueger, A.B. (1992) The effect of minimum wage on the fast food industry. *Industrial and Labor Relations Review*, 46(1), 6–21. available from: <https://doi.org/10.1177/001979399204600102>
- Manson, S., Schroeder, J., Riper, D.V., Knowles, K., Kugler, T., Roberts, F., et al. (2022) *IPUMS National Historical Geographic Information System: Version 18.0*. Minneapolis. available from: <https://doi.org/10.18128/D050.V18.0>
- Miller, S., Wagner, G. & Plemmons, J. (2025) *ECIN replication package for "Employment effects of minimum wage after indexing: establishment evidence from Oregon restaurants"*. nnrbor: Inter-University Consortium for Political and Social Research [distributor] 2025-01-01. available from: <https://doi.org/10.3886/E208522V4>

- Nathanson, C.G. & Zwick, E. (2018) Restored development: theory and evidence of supply-side speculation in the housing market. *The Journal of Finance*, 73(6), 2587–2633. available from: <https://doi.org/10.1111/jofi.12719>
- Neumark, D. & Kolko, J. (2010) Do enterprise zones create jobs? Evidence from California's enterprise zone program. *Journal of Urban Economics*, 68(1), 1–19. ISSN: 0094-1190. available from: <https://doi.org/10.1016/j.jue.2010.01.002>
- Neumark, D., Wall, B. & Zhang, J. (2011) Do small businesses create more jobs? New evidence for the United States from the National Establishment Time Series. *The Review of Economics and Statistics*, 93(1), 16–29. ISSN: 0034-6535. available from: https://doi.org/10.1162/REST_a_00060
- Neumark, D. & Wascher, W. (2000) Minimum wage and employment: a case study of the fast-food industry in New Jersey and Pennsylvania: comment. *The American Economic Review*, 90(5), 1362–1396. available from: <https://doi.org/10.1257/aer.90.5.1362>
- Neumark, D., Zhang, J. & Wall, B. (2007) Employment dynamics and business relocation: new evidence from the National Establishment Time Series. *Research in Labor Economics*, 26, 39–83. available from: [https://doi.org/10.1016/s0147-9121\(06\)26002-3](https://doi.org/10.1016/s0147-9121(06)26002-3)
- Pérez, J. (2022) City minimum wages and spatial equilibrium effects. Working paper.
- Persky, J. & Baiman, R. (2010) Do state minimum wage laws reduce employment? Mixed messages from fast food outlets in Illinois and Indiana. *Journal of Regional Analysis and Policy*, 40(2), 132–142.
- Powell, D. (2022) Synthetic control estimation beyond comparative case studies: does the minimum wage reduce employment? *Journal of Business & Economic Statistics*, 40(3), 1302–1314. available from: <https://doi.org/10.1080/07350015.2021.1927743>
- Powers, E.T. (2009) The impact of minimum-wage increases: evidence from fast-food establishments in Illinois and Indiana. *Journal of Labor Research*, 30(4), 365–394. available from: <https://doi.org/10.1007/s12122-009-9068-3>
- Powers, E.T., Baiman, R. & Persky, J. (2007) *Impacts of the Illinois minimum wage on the fast food industry: a report to the Russell Sage Foundation*. Champaign, IL: University of Illinois at Urbana–Champaign.
- Rosenbaum, P.R. (2020) Modern algorithms for matching in observational studies. *Annual Review of Statistics and Its Application*, 7(1), 143–176. available from: <https://doi.org/10.1146/annurev-statistics-031219-041058>
- Rosenbaum, P.R. & Rubin, D.B. (1983) The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55. available from: <https://doi.org/10.1093/biomet/70.1.41>
- Rossi-Hansberg, E., Sarte, P.-D. & Trachter, N. (2018) *Diverging trends in national and local concentration*. Working Paper 25066. National Bureau of Economic Research. available from: <https://doi.org/10.3386/w25066>
- Rubin, D.B. (1973) The use of matched sampling and regression adjustment to remove bias in observational studies. *Biometrics*, 29(1), 185–203. available from: <https://doi.org/10.2307/2529685>
- Rubin, D.B. (1979) Using multivariate matched sampling and regression adjustment to control bias in observational studies. *Journal of the American Statistical Association*, 74(366a), 318–328. available from: <https://doi.org/10.1080/01621459.1979.10482513>
- Rubin, D.B. & Thomas, N. (2000) Combining propensity score matching with additional adjustments for prognostic covariates. *Journal of the American Statistical Association*, 95(450), 573–585. available from: <https://doi.org/10.1080/01621459.2000.10474233>
- Sant'Anna, P.H.C. & Zhao, J. (2020) Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1), 101–122. ISSN: 0304-4076. available from: <https://doi.org/10.1016/j.jeconom.2020.06.003>
- Singell, L.D. & Terborg, J.D. (2007) Employment effects of two northwest minimum wage initiatives. *Economic Inquiry*, 45(1), 40–55. available from: <https://doi.org/10.1111/j.1465-7295.2006.00018.x>
- Walls & Associates. (2019) National Establishment Time Series.
- Zeballos, E. & Marchesi, K. (2022) Comparing food sector employment headcount and sales data in the National Establishment Time Series Database to Federal Data. USD Economic Research Service Technical Bulletin Number 1958.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Miller, S., Wagner, G. & Plemmons, . (2025) Employment effects of minimum wage indexing: establishment evidence from Oregon restaurants. *Economic Inquiry*, 1–34. available from: <https://doi.org/10.1111/ecin.13284>

PPENDIX

EVENT STUDY PLOTS FOR ROBUSTNESS CHECKS

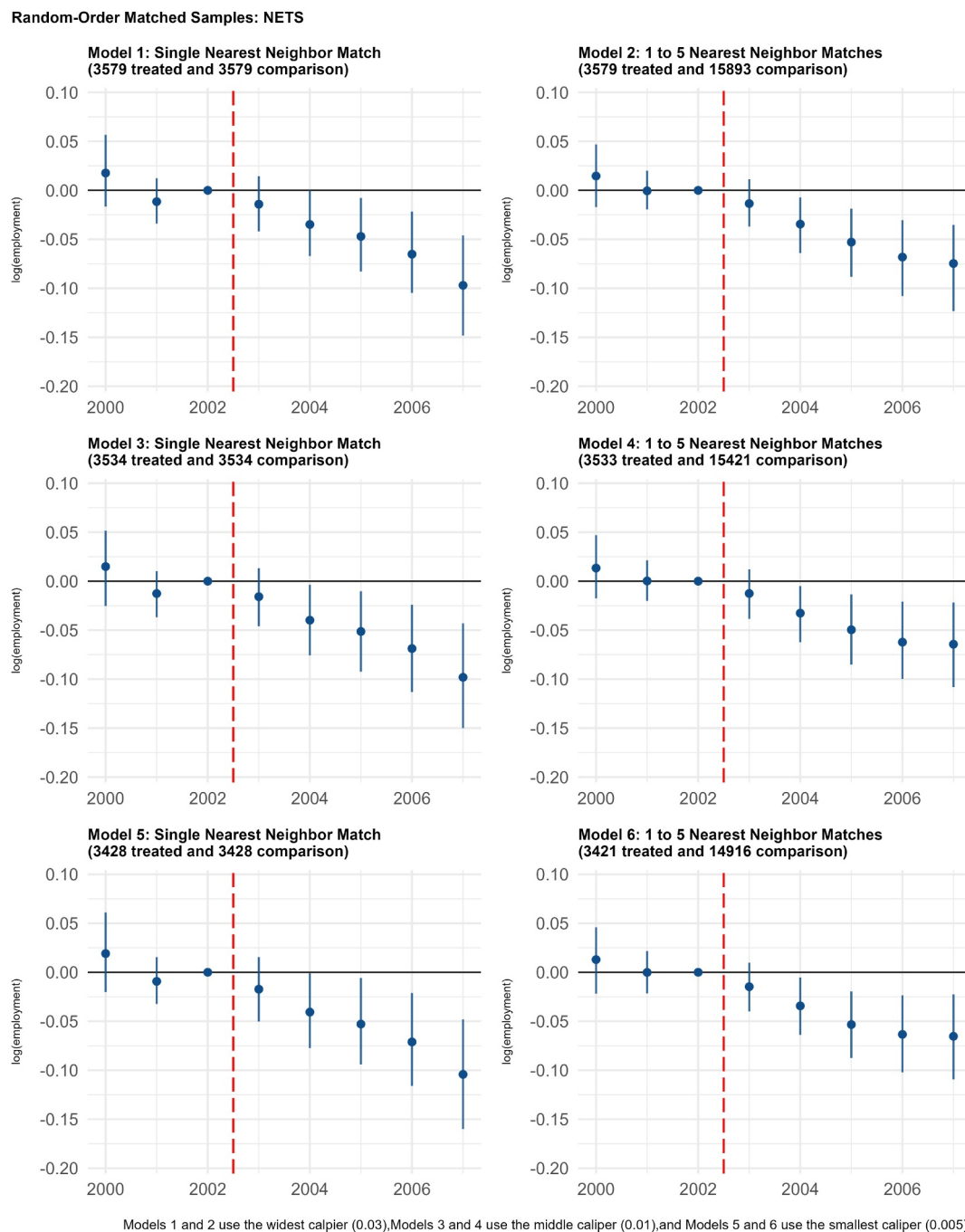


FIGURE 1 Event study plots: random-order matching using NETS database. Each panel shows the event study version of the matched sample regressions presented in Table 3. Model numbers in the figure correspond to the model numbers presented in Table 3. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). NETS, National Establishment Time Series.

Random-Order Matched Samples: YTS

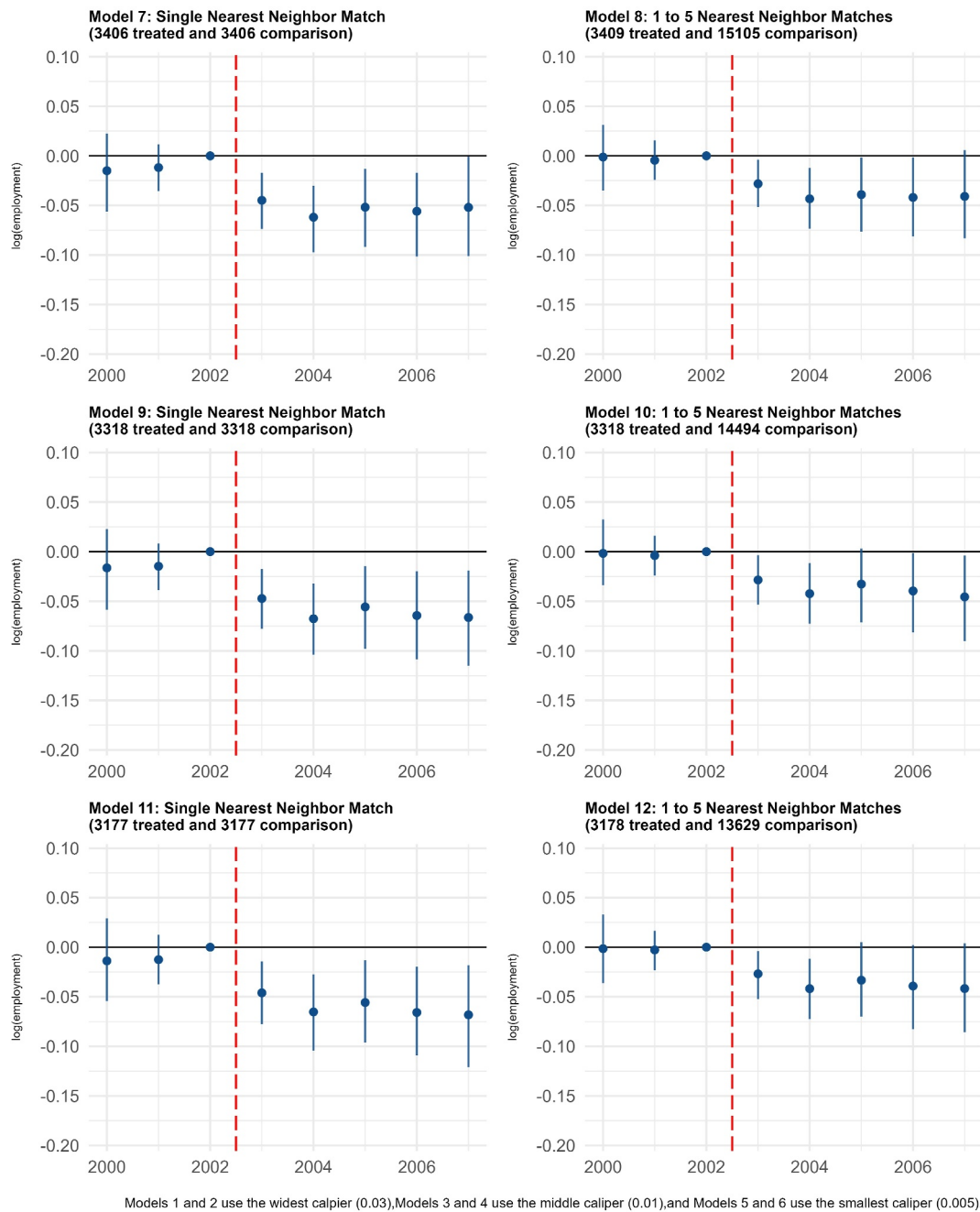
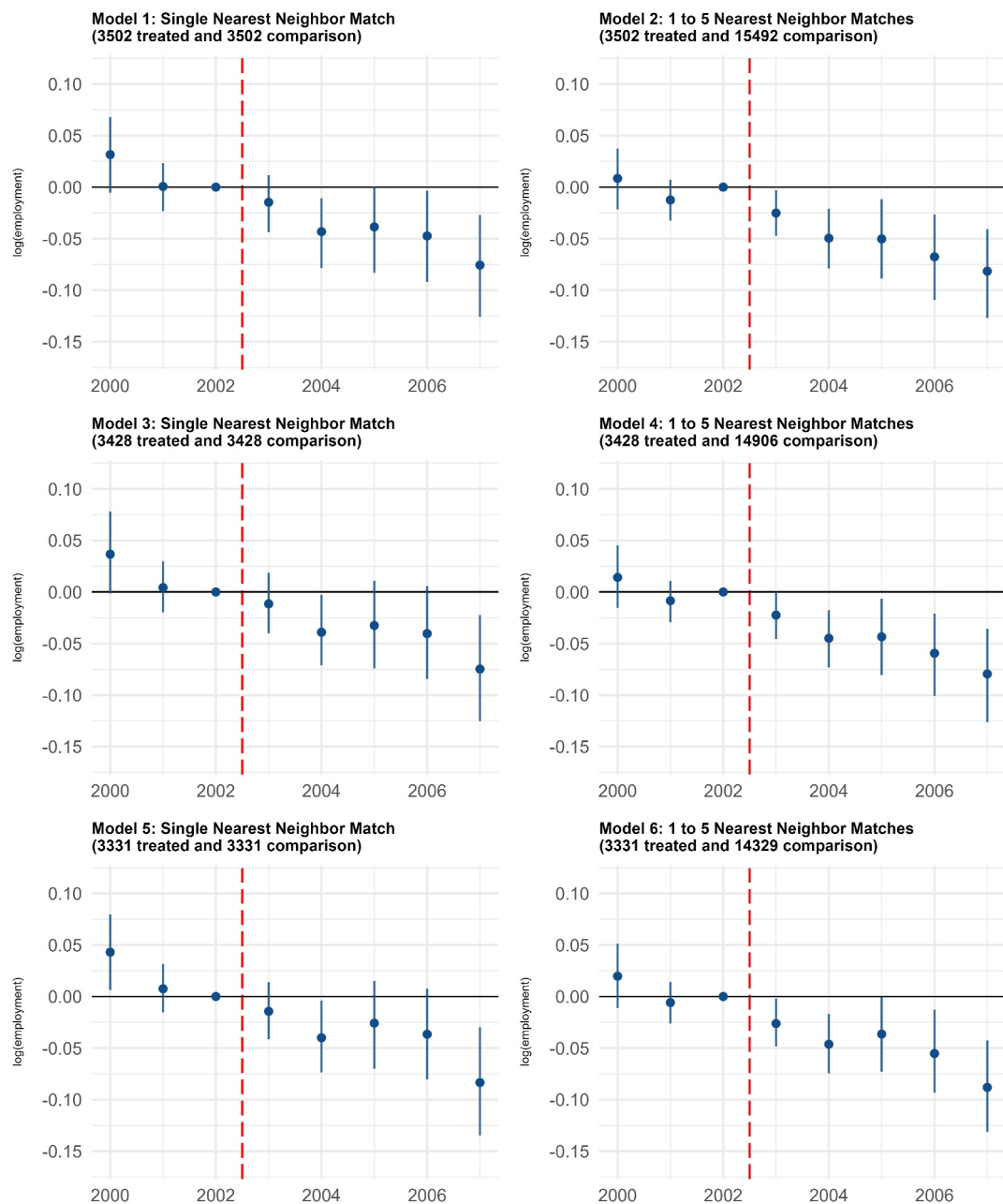


FIGURE 2 Event study plots: random-order matching using YTS database. Each panel shows the event study version of the matched sample regressions presented in Table 3. Model numbers in the figure correspond to the model numbers presented in Table 3. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). YTS, Your Economy Time Series database.

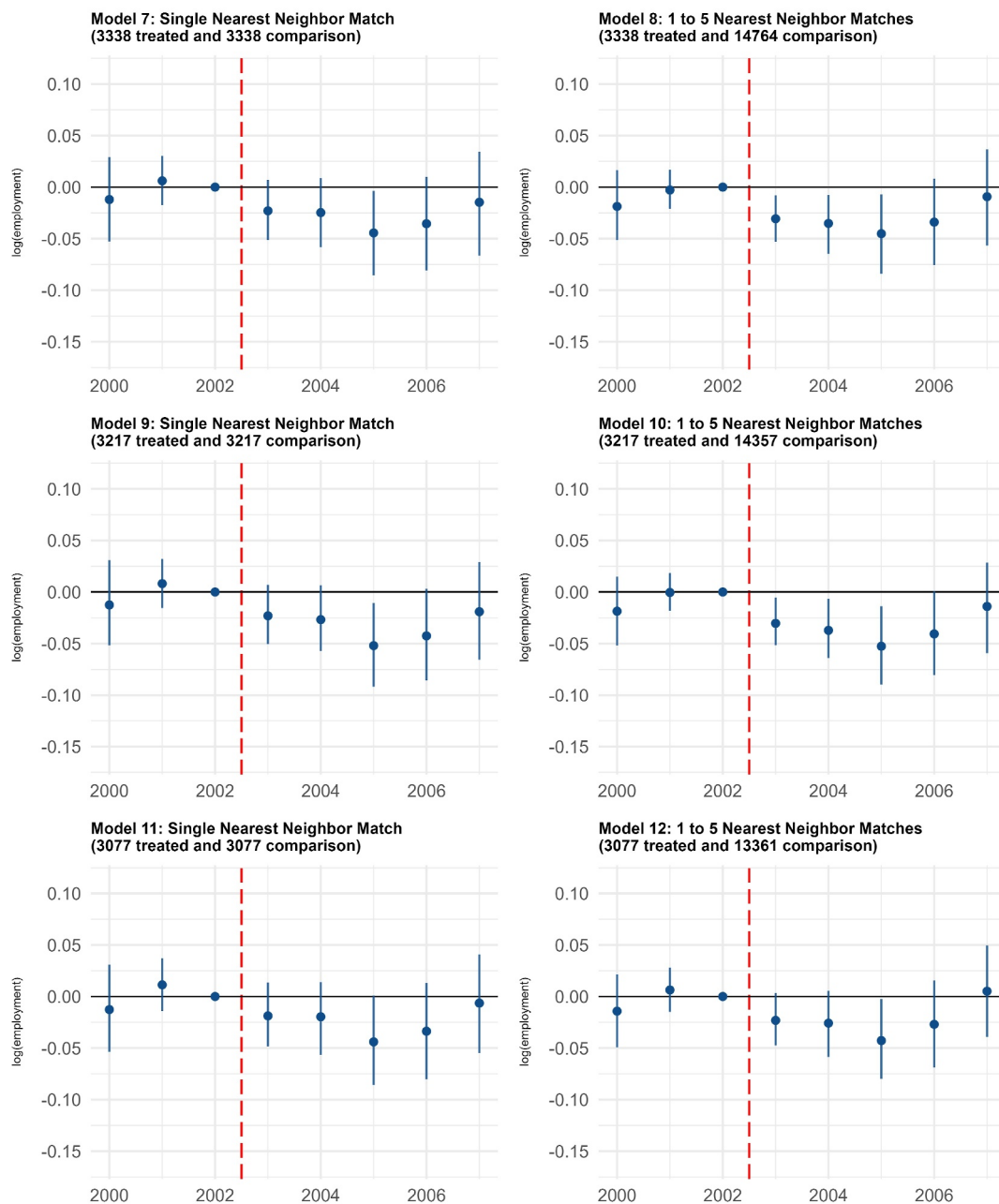
Matched Samples Excluding Non-Employer Establishments: NETS



Models 1 and 2 use the widest caliper (0.03), Models 3 and 4 use the middle caliper (0.01), and Models 5 and 6 use the smallest caliper (0.005).

FIGURE 3 Event study plots: non-employer establishments dropped, NETS. Each panel shows the event study version of the matched sample regressions presented in Table 4. Model numbers in the figure correspond to the model numbers presented in Table 4. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). NETS, National Establishment Time Series.

Matched Samples Excluding Non-Employer Establishments: YTS



Models 1 and 2 use the widest caliper (0.03), Models 3 and 4 use the middle caliper (0.01), and Models 5 and 6 use the smallest caliper (0.005).

FIGURE 4 Event study plots: non-employer establishments dropped, YTS. Each panel shows the event study version of the matched sample regressions presented in Table 4. Model numbers in the figure correspond to the model numbers presented in Table 4. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). YTS, Your Economy Time Series database.

Matched Samples Excluding Small Establishments: NETS

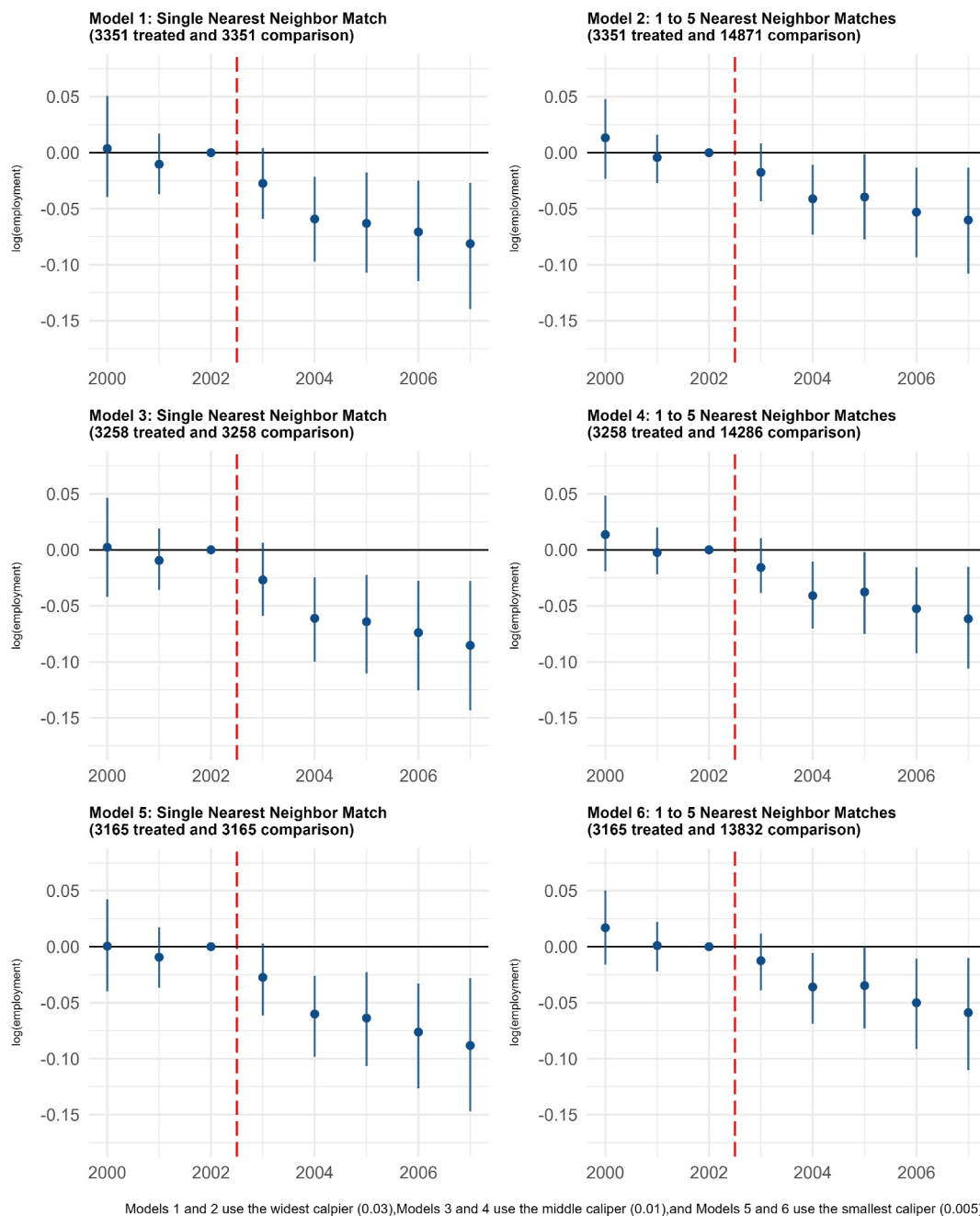
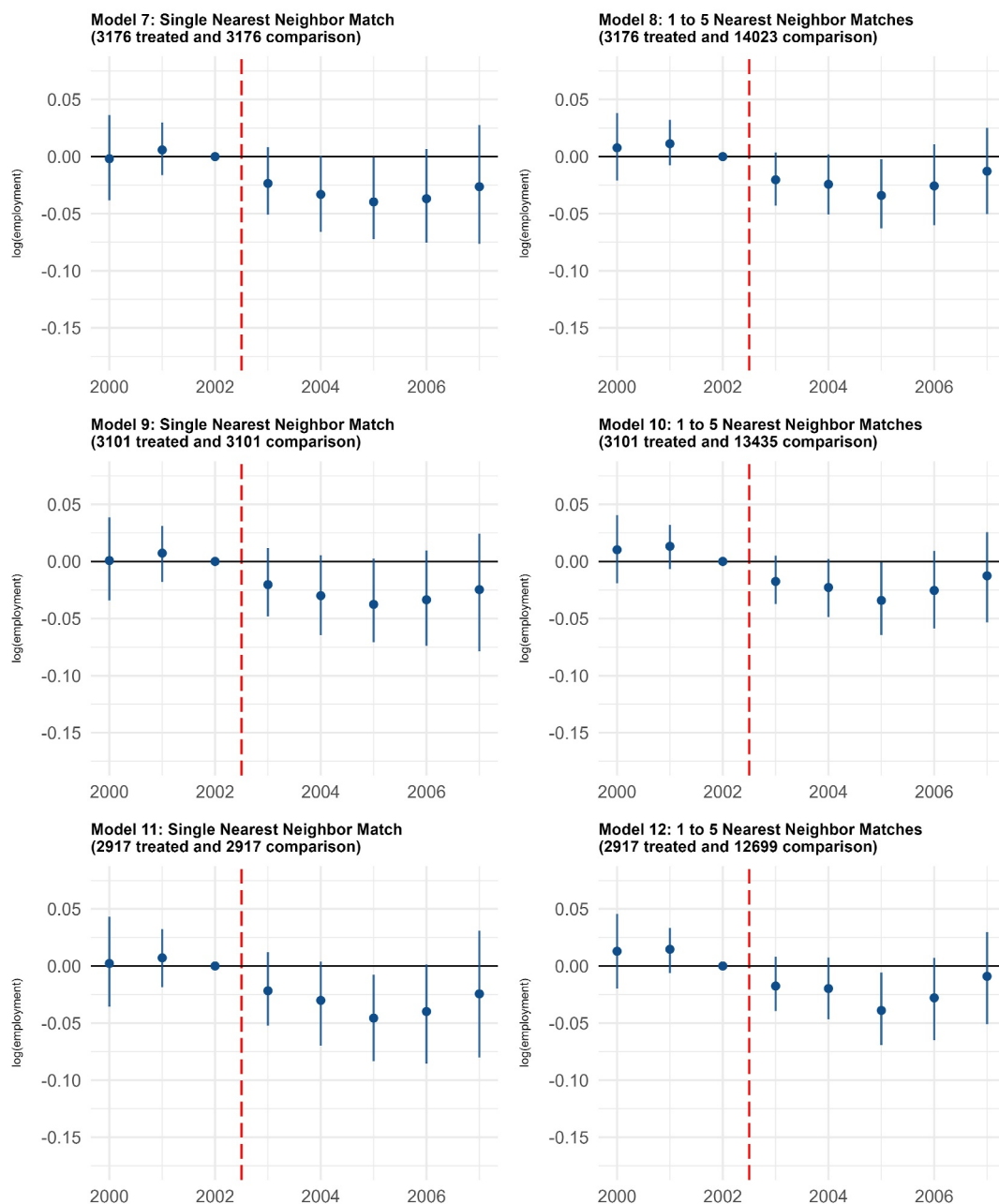


FIGURE 5 Event study plots: small establishments dropped, NETS. Each panel shows the event study version of the matched sample regressions presented in Table 5. Model numbers in the figure correspond to the model numbers presented in Table 5. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following badie and Spiess (2022). NETS, National Establishment Time Series.

Matched Samples Excluding Small Establishments: YTS



Models 1 and 2 use the widest caliper (0.03), Models 3 and 4 use the middle caliper (0.01), and Models 5 and 6 use the smallest caliper (0.005).

FIGURE 6 Event study plots: small establishments dropped, YTS. Each panel shows the event study version of the matched sample regressions presented in Table 5. Model numbers in the figure correspond to the model numbers presented in Table 5. Error bars show the 95% confidence intervals from 1000 blocked bootstrap replications sampling on matched sets following [badie and Spiess \(2022\)](#). YTS, Your Economy Time Series database.